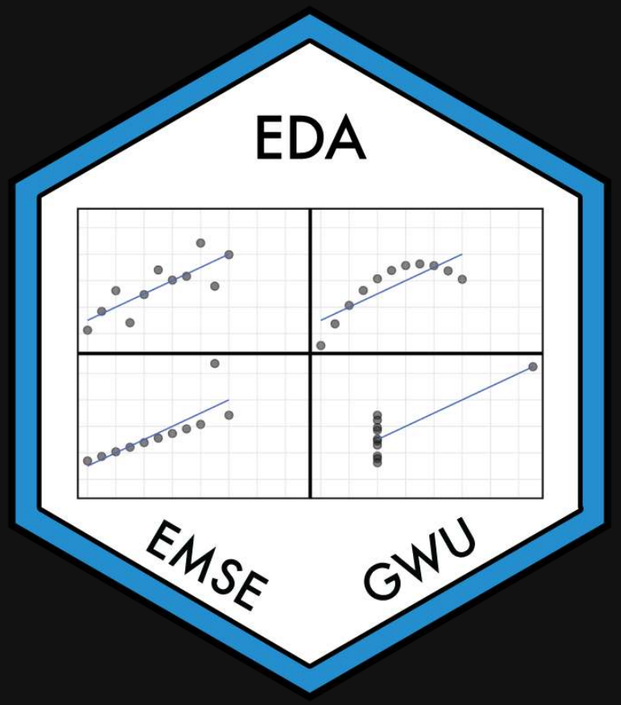


Trends

EMSE 4572/6572: Exploratory Data Analysis

John Paul Helveston

October 28, 2026

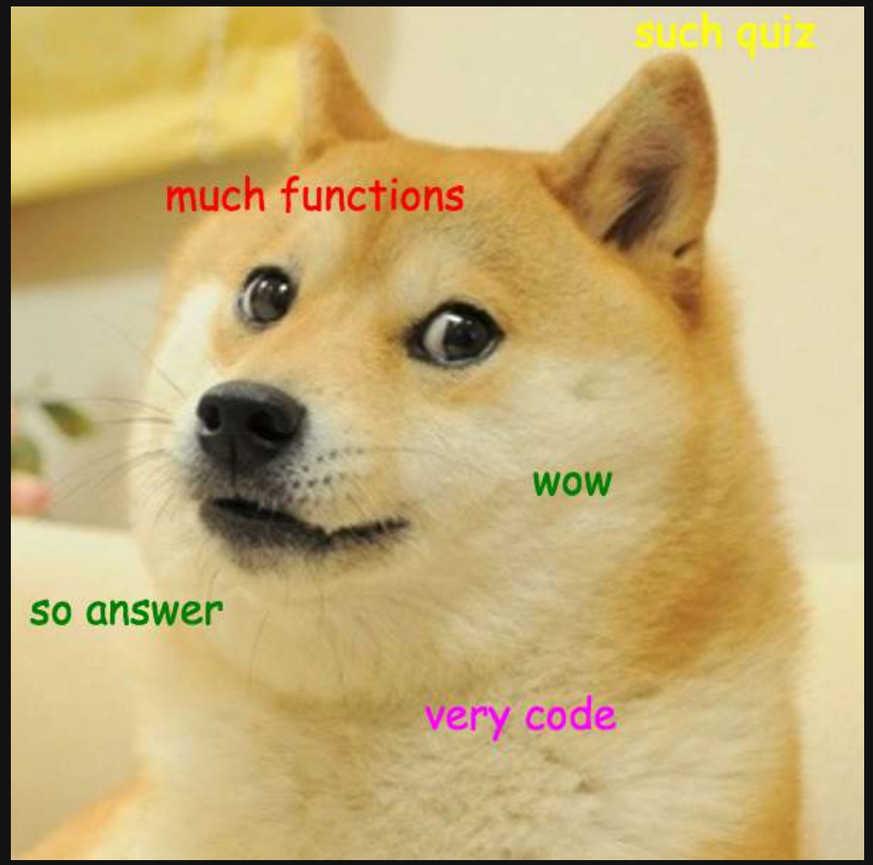
Quiz 4

12:00

Write your name on the quiz!

Rules:

- Work alone; no outside help of any kind is allowed.
- No calculators, no notes, no books, no computers, no phones.



Today's data

Seen before:

```
gapminder <- read_csv(file.path('data', 'gapminder.csv'))
milk_production <- read_csv(file.path('data', 'milk_production.csv'))
global_temps <- read_csv(file.path('data', 'nasa_global_temps.csv'))
internet_region <- read_csv(file.path('data', 'internet_users_region.csv'))
hotdogs <- read_csv(file.path('data', 'hot_dog_winners.csv'))
internet_country <- read_csv(file.path('data', 'internet_users_country.csv'))
```

New datasets:

```
us_covid <- read_csv(file.path('data', 'us_covid.csv'))
us_diseases <- read_csv(file.path('data', 'us_contagious_diseases.csv'))
```

New packages:

```
1 install.packages('viridis')
2 install.packages('gganimate')
3 install.packages('magick')
```

Week 10: *Trends*

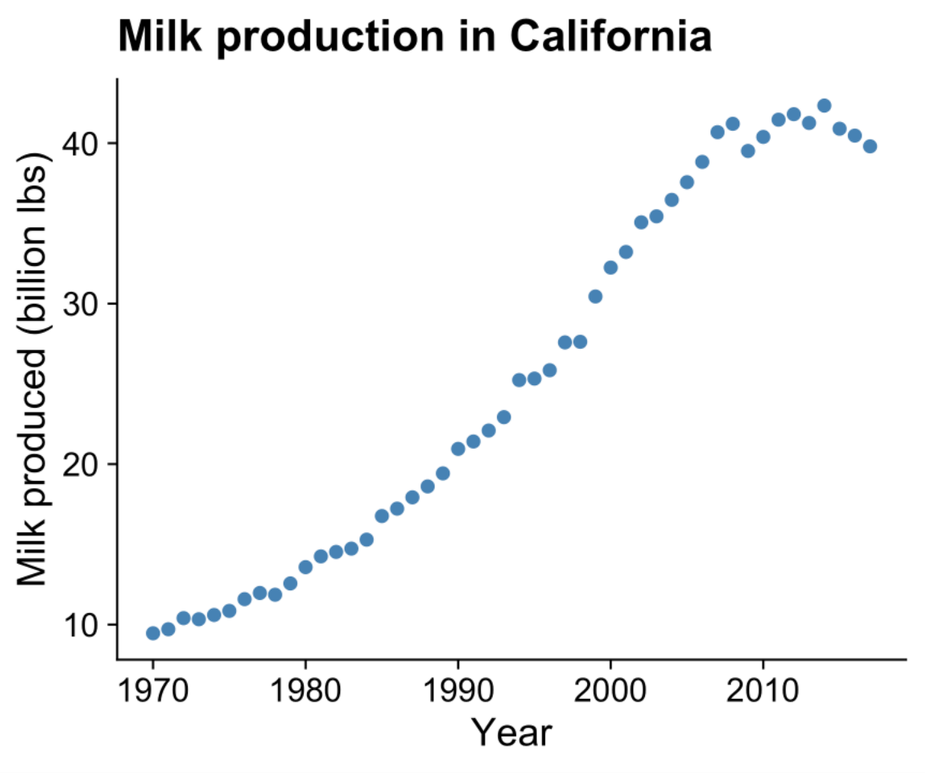
1. Single Variables
2. Animations
3. Multiple Variables

Week 10: *Trends*

1. Single Variables
2. Animations
3. Multiple Variables

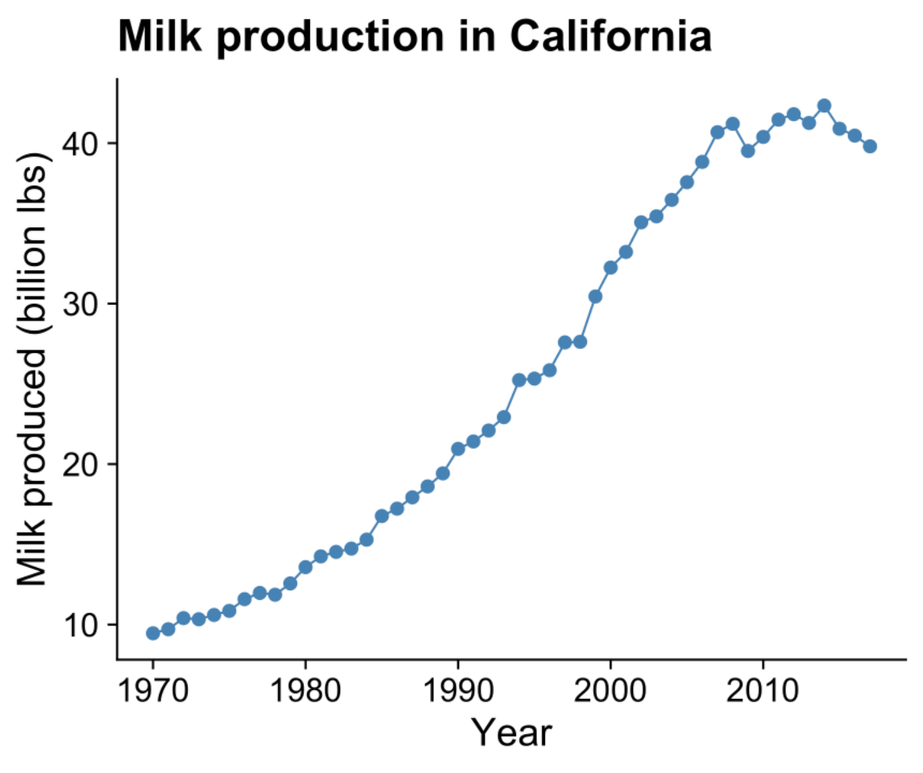
Points

Plotting the data points is a good starting point for viewing trends



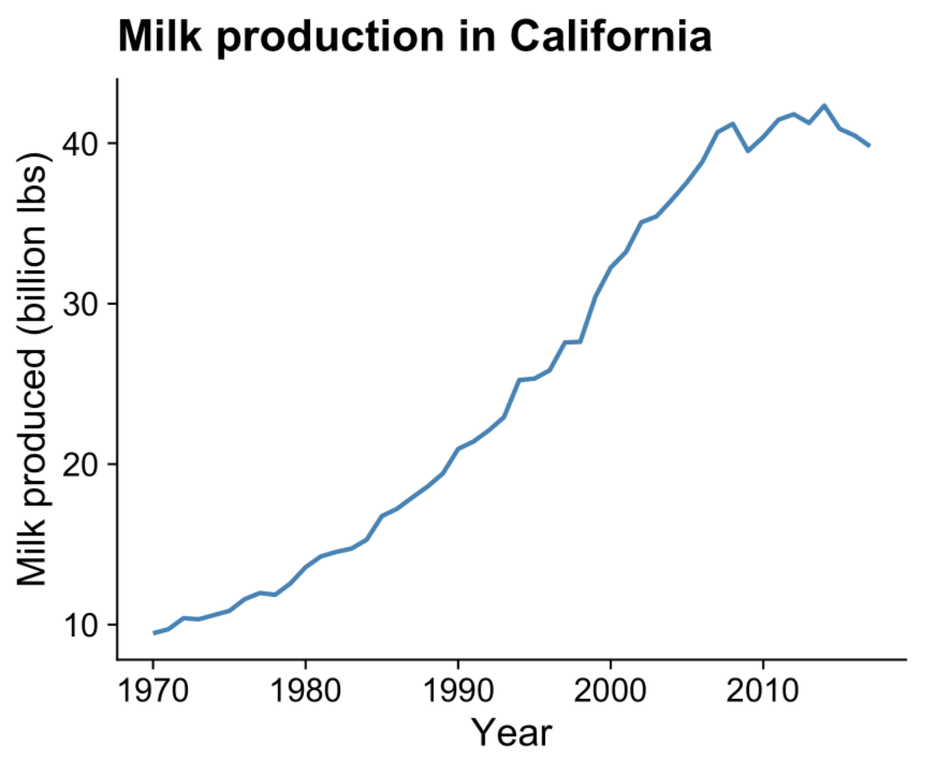
Points + line

Adding lines between the points helps see the overall trend



Line

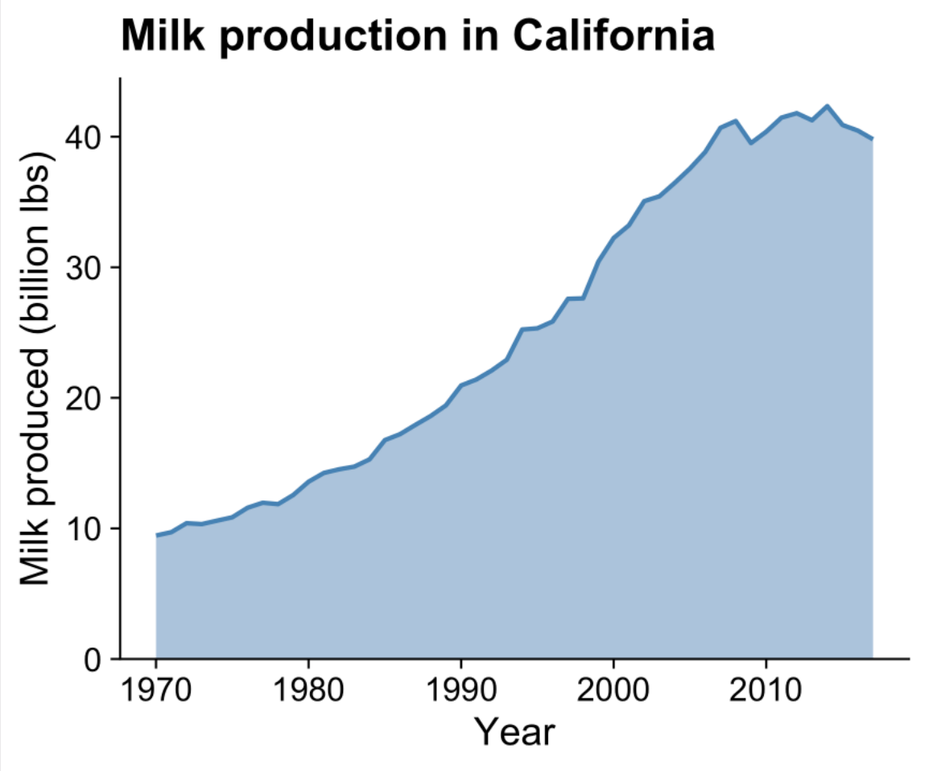
Omitting the points emphasizes the overall trend



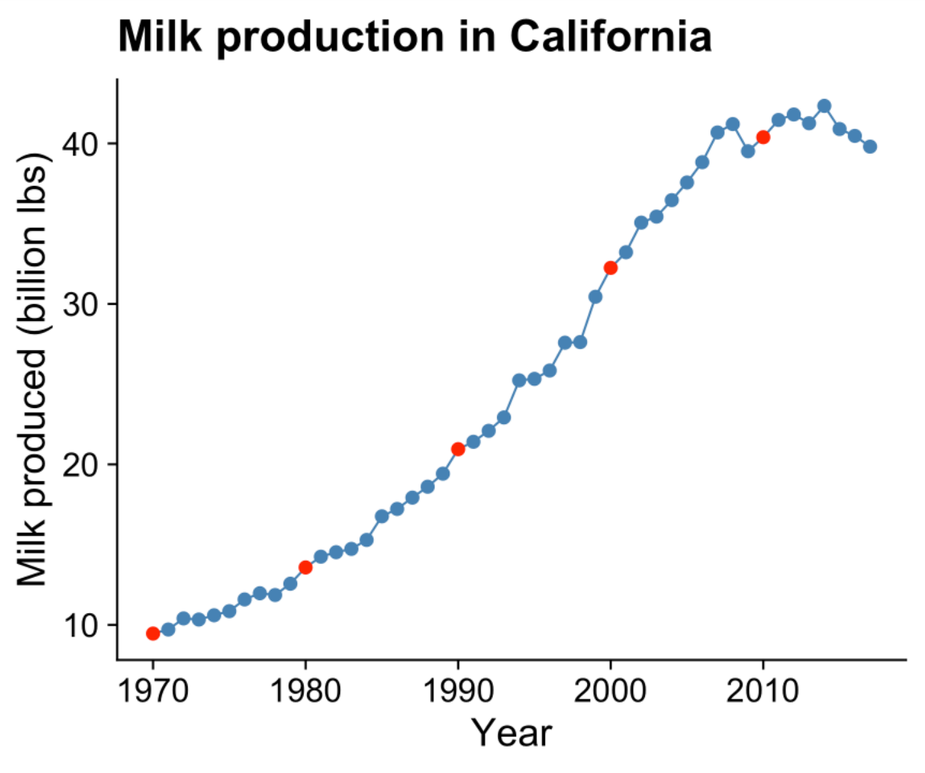
Line + area

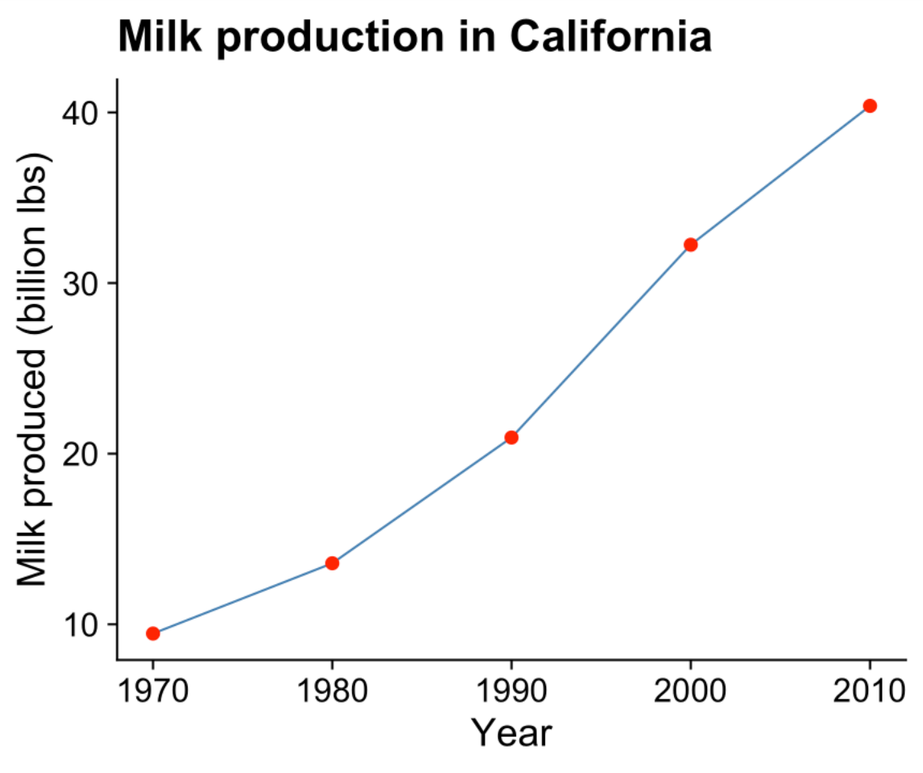
Filling area below line emphasizes cumulative over time

(y-axis should start at 0)



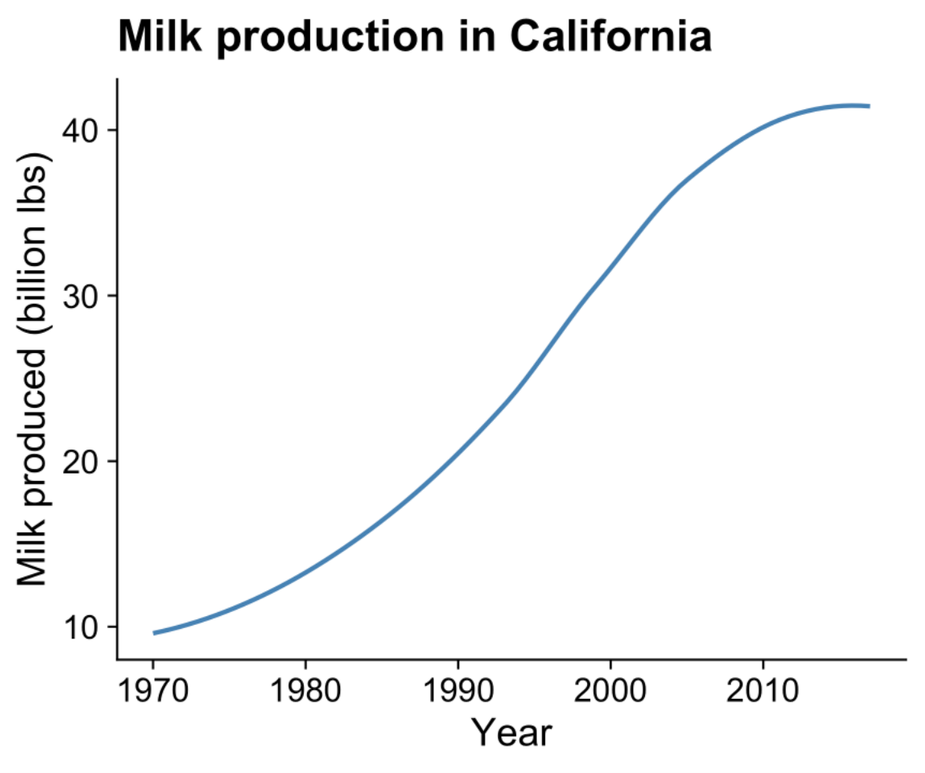
If points are too sparse, a line can be misleading





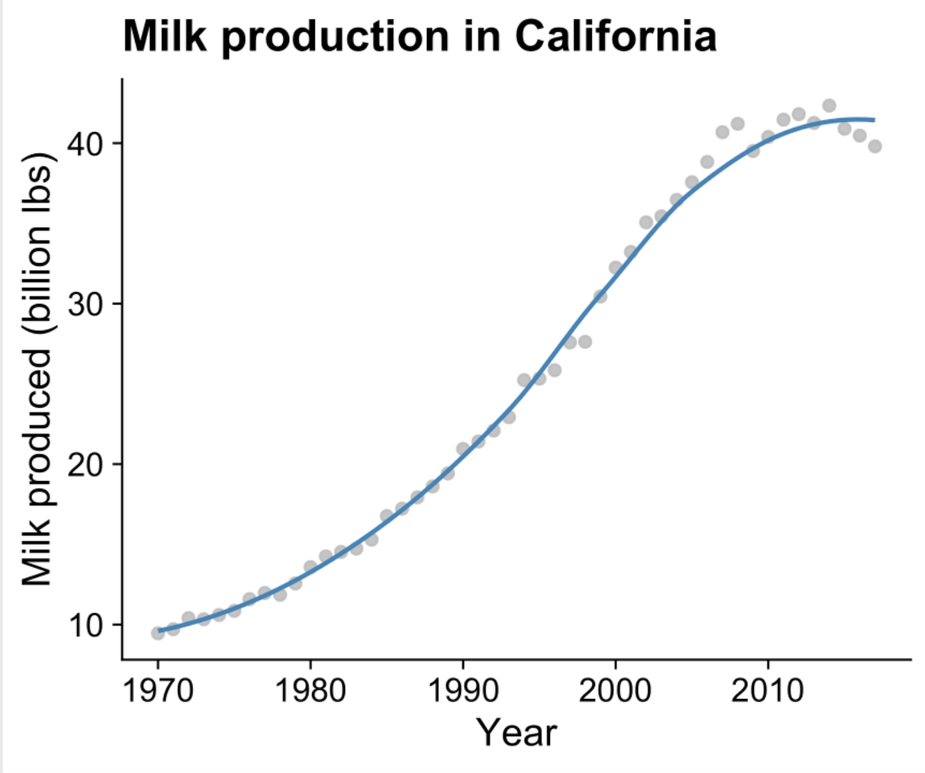
Smoothed line

Adding a “smoothed” line shows a modeled representation of the overall trend

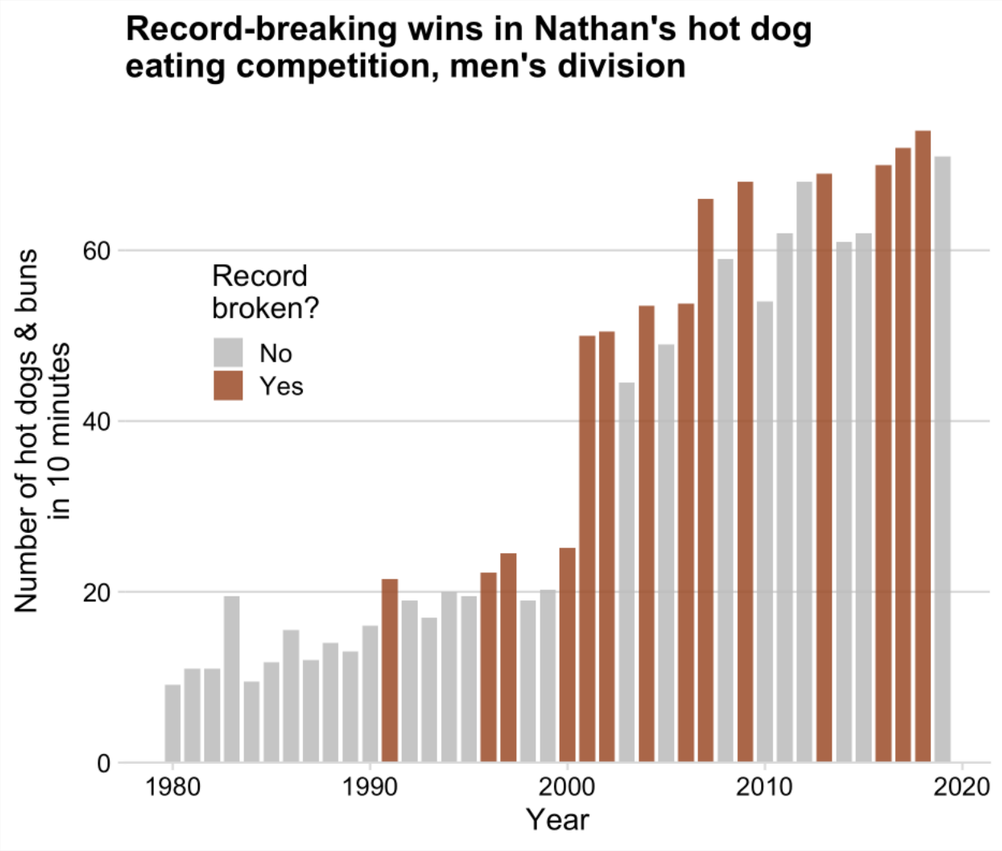


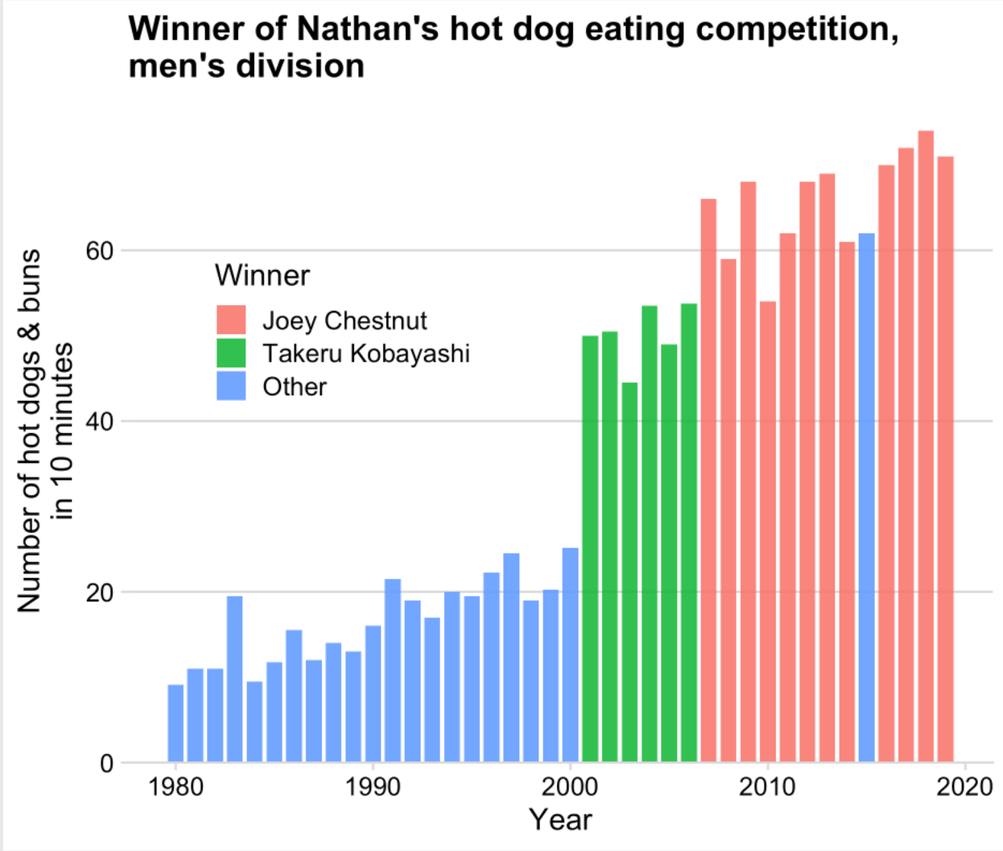
Smoothed line + points

Putting the smoothed line over the data points helps show whether **outliers** are driving the trend line



Bars are useful when emphasizing the **data points** rather than the **slope between them**

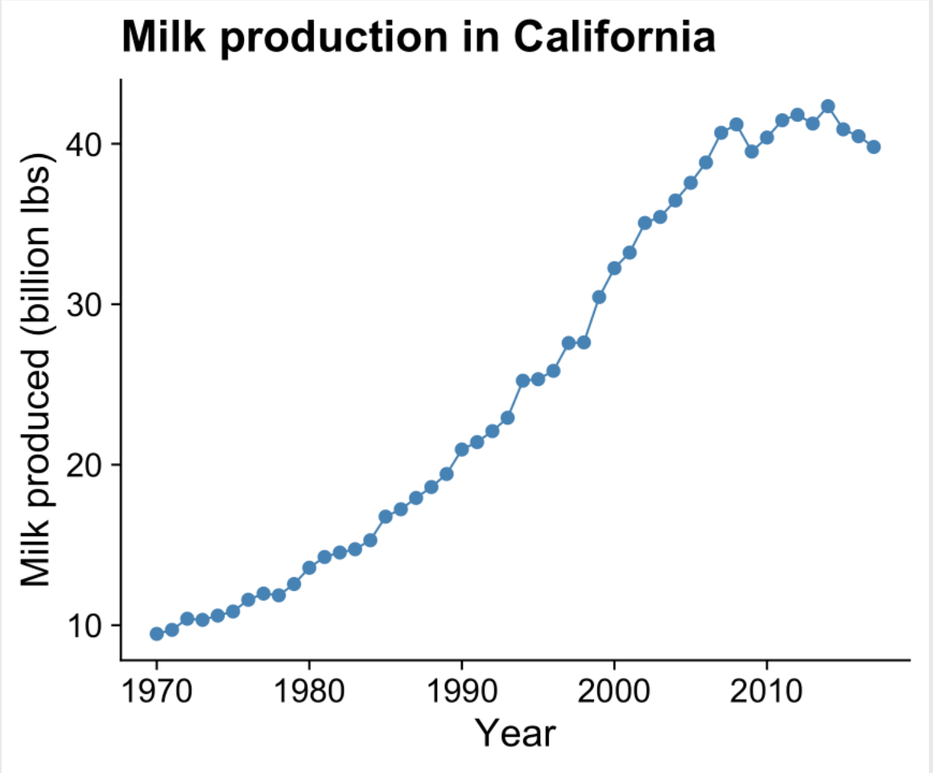




How to: **Points + line**

Be sure to draw the line first,
then overlay the points

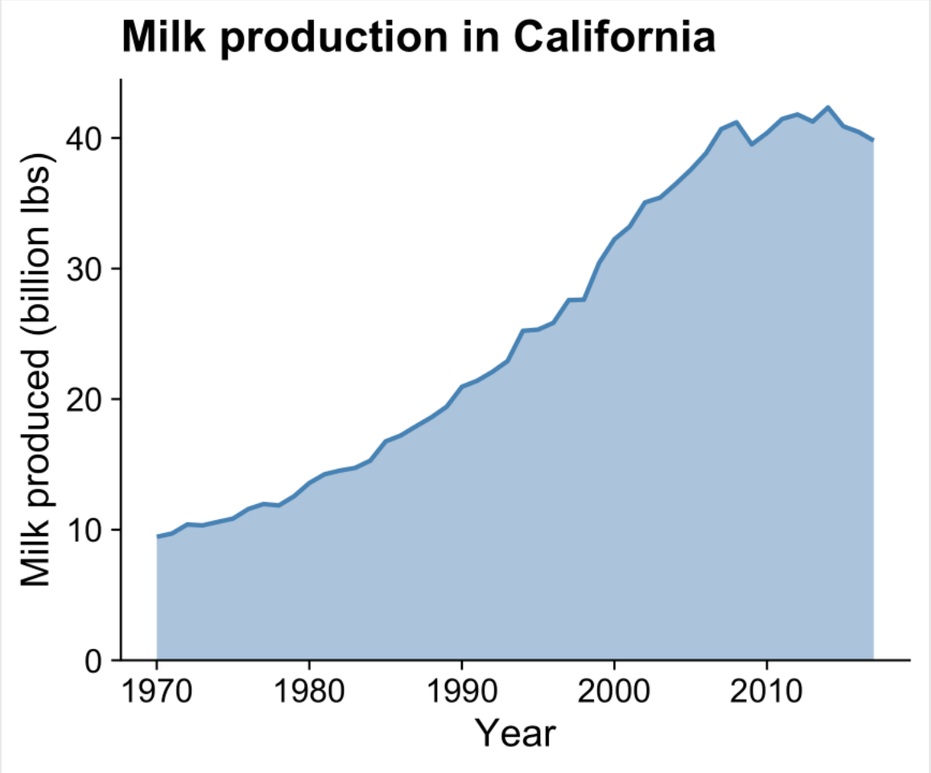
```
ggplot(milk_ca, aes(x = year, y = milk_produced)) +
    geom_line(color = 'steelblue', size = 0.5) +
    geom_point(color = 'steelblue', size = 2) +
    theme_half_open(font_size = 18) +
    labs(
        x = 'Year',
        y = 'Milk produced (billion lbs)',
        title = 'Milk production in California'
    )
```



How to: **Line + area**

Likewise, draw the area first then overlay the line

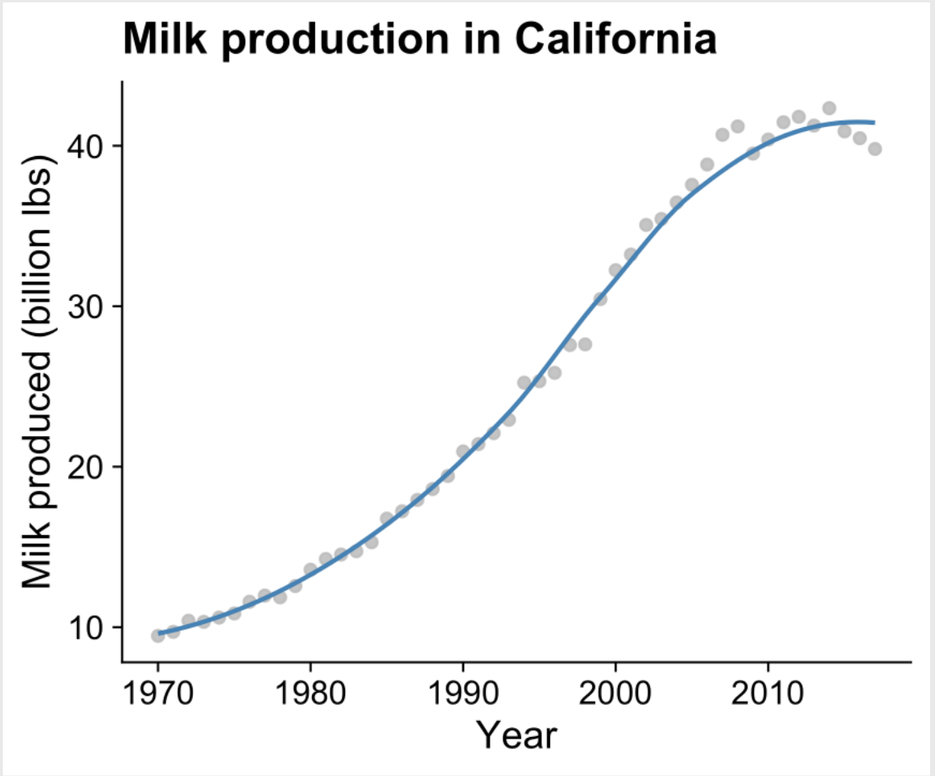
```
ggplot(milk_ca, aes(x = year, y = milk_produced)) +
    geom_area(fill = 'steelblue', alpha = 0.5) +
    geom_line(color = 'steelblue', size = 1) +
    scale_y_continuous(
        expand = expansion(mult = c(0, 0.05))
    ) +
    theme_half_open(font_size = 18) +
    labs(
        x = 'Year',
        y = 'Milk produced (billion lbs)',
        title = 'Milk production in California'
    )
```



How to: **Smoothed line + points**

Use `alpha` to make points slightly transparent

```
ggplot(milk_ca, aes(x = year, y = milk_produced)) +
  geom_point(color = 'grey', size = 2, alpha = 0.9)
  geom_smooth(color = 'steelblue', size = 1, se = FA
  theme_half_open(font_size = 18) +
  labs(
    x = 'Year',
    y = 'Milk produced (billion lbs)',
    title = 'Milk production in California'
  )
```



Your turn

15:00

Use the `global_temps` data frame to explore ways to visualize the change in average global temperatures.

Consider using:

- points
- lines
- areas
- smoothed lines

```
1 global_temps <- read_csv(file.path(
2   'data',
3   'nasa_global_temps.csv'
4 ))
5
6 head(global_temps)
```

```
#> # A tibble: 6 × 3
#>    year meanTemp smoothTemp
#>   <dbl>    <dbl>      <dbl>
#> 1  1880    -0.15      -0.08
#> 2  1881    -0.07      -0.12
#> 3  1882    -0.1       -0.15
#> 4  1883    -0.16      -0.19
#> 5  1884    -0.27      -0.23
#> 6  1885    -0.32      -0.25
```

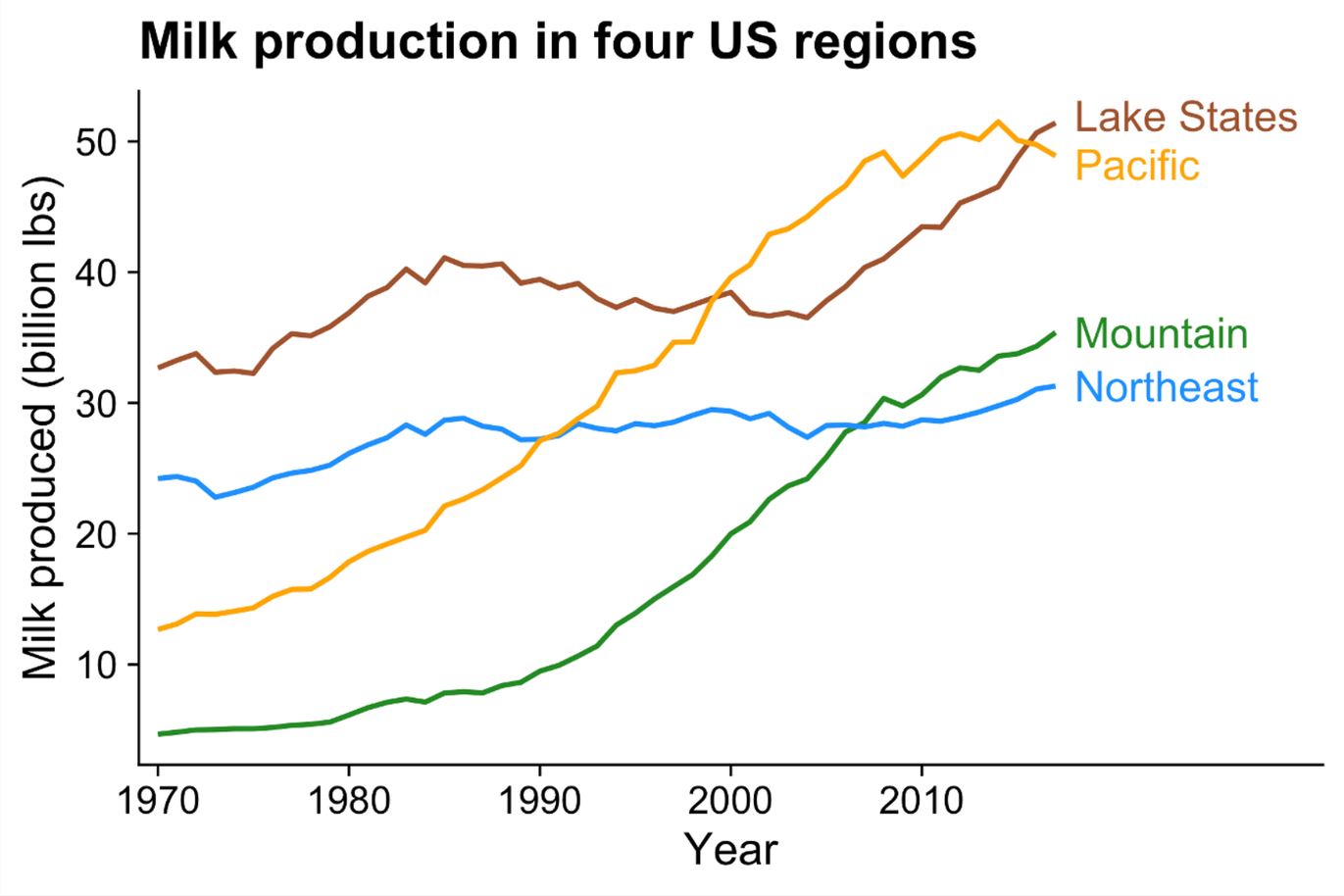
Week 10: *Trends*

1. Single Variables
2. Animations
3. Multiple Variables

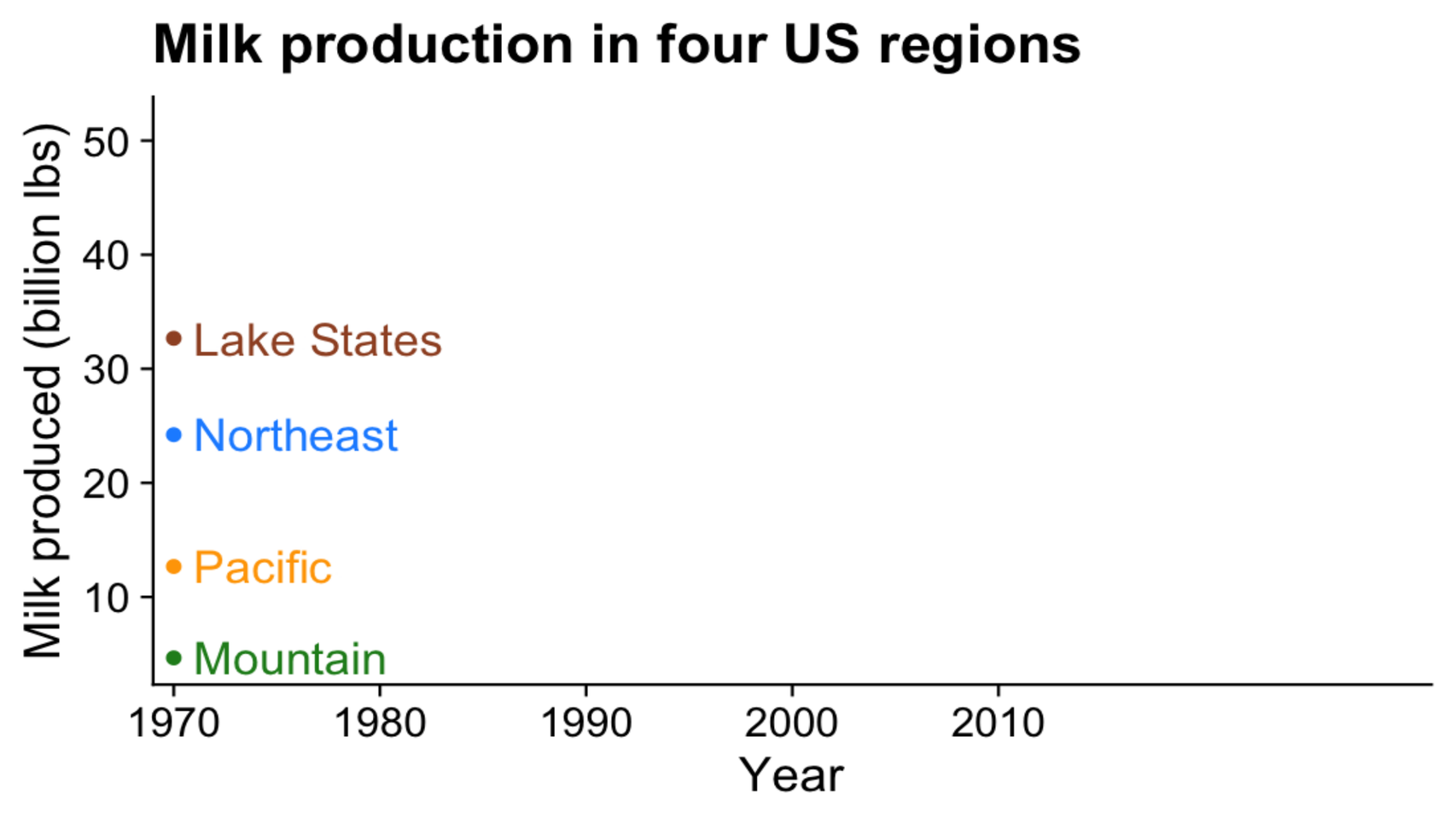
Animation adds emphasis to the **change over time**

...plus it's fun!

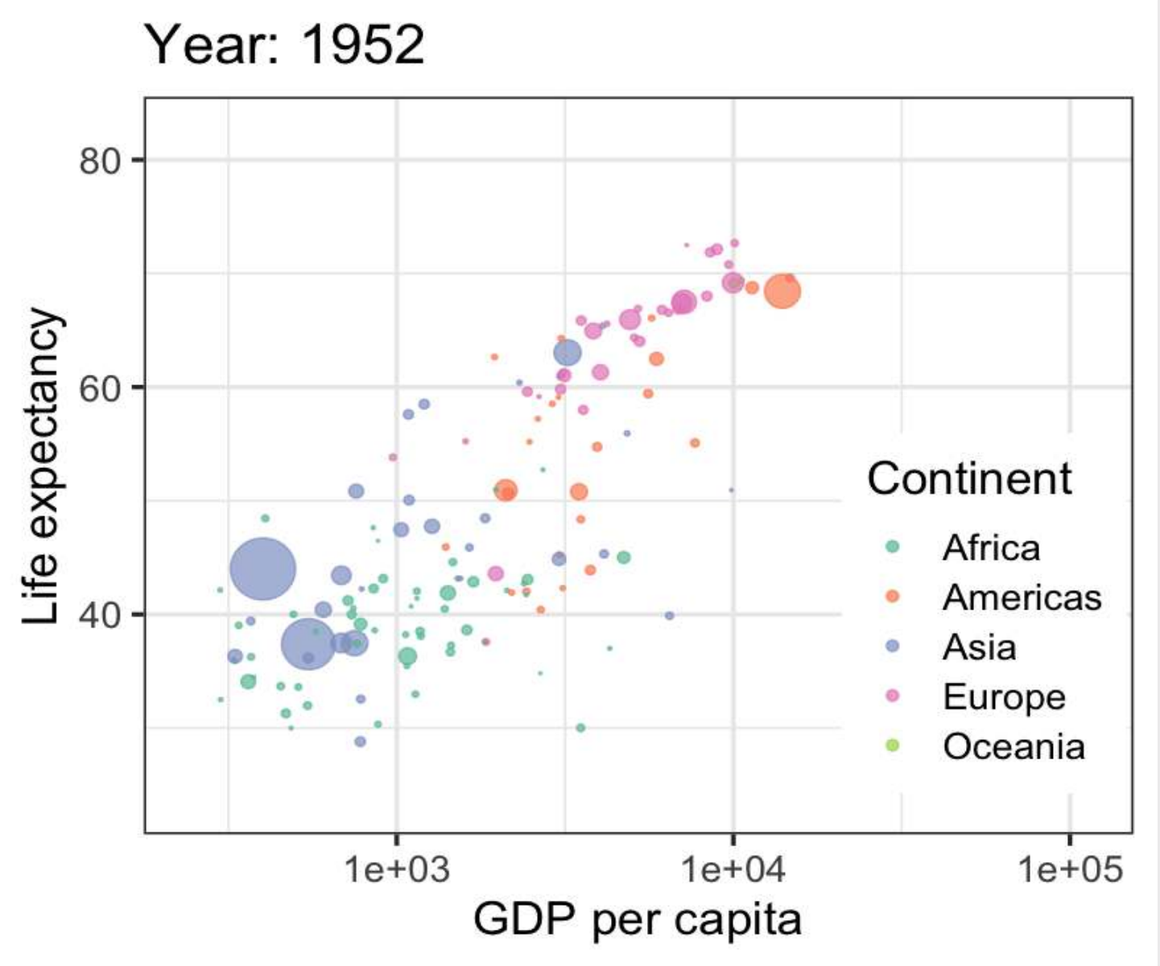
Static chart



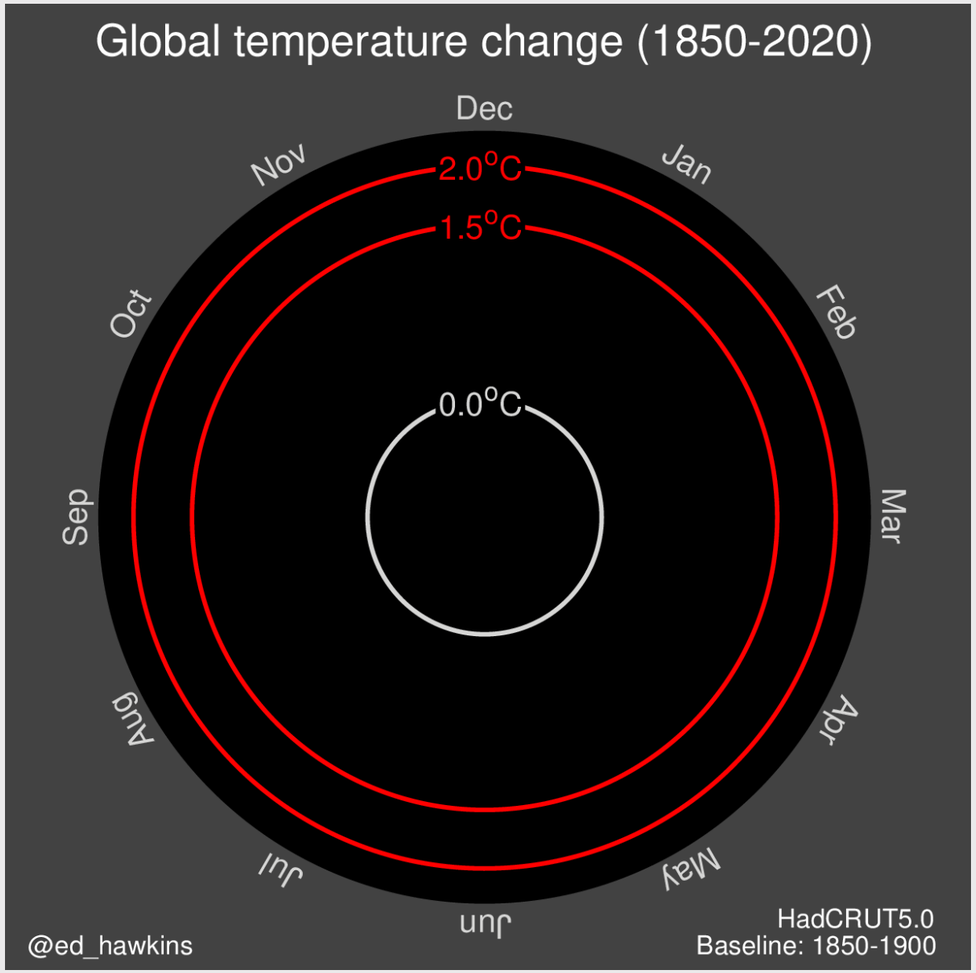
Animated chart



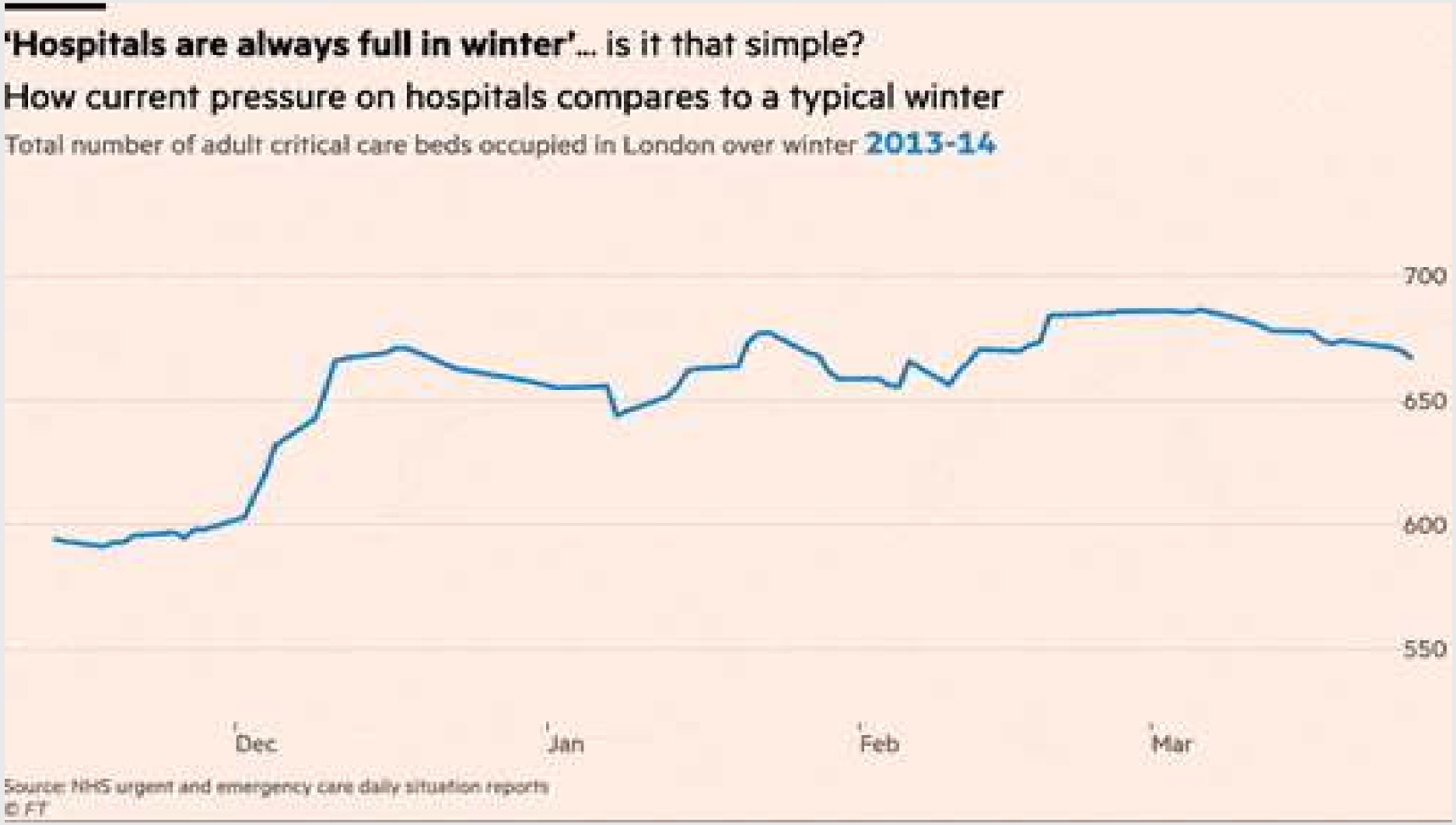
Animation is particularly helpful for the **time dimension**



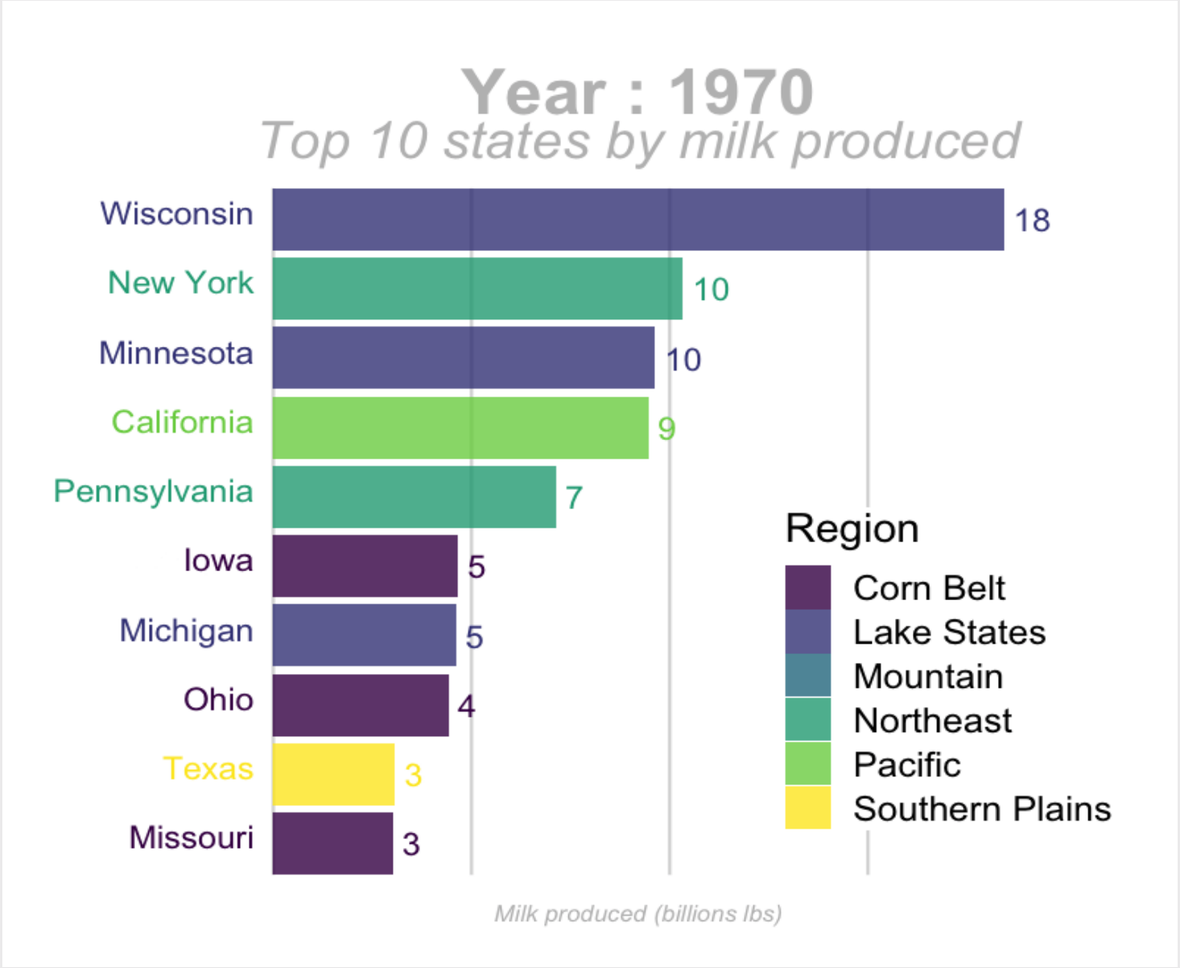
Animation is particularly helpful for the **time dimension**



Animation is particularly helpful for the **time dimension**



Animation is particularly helpful for the **time dimension**

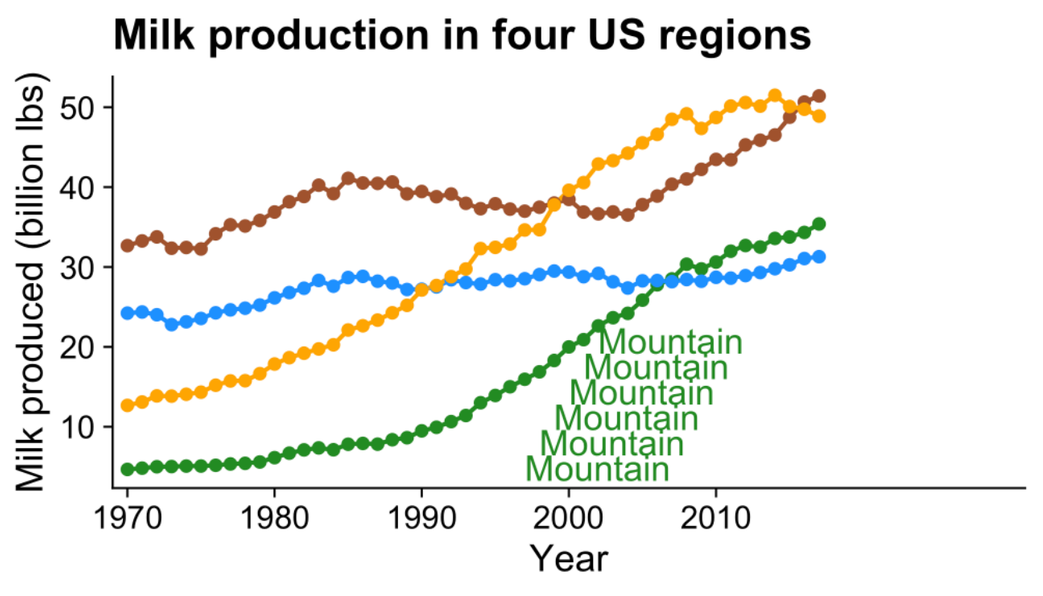


How to: **Animate a line plot**

Make a static plot w/labels for each year

```
milk_region_anim_plot <- milk_region %>%
  ggplot(aes(x = year, y = milk_produced, color = region)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  geom_text_repel(
    aes(label = region),
    hjust = 0,
    nudge_x = 1,
    direction = "y",
    size = 6,
    segment.color = NA
  ) +
  scale_x_continuous(
    breaks = seq(1970, 2010, 10),
    expand = expansion(add = c(1, 13))
  ) +
  scale_color_manual(
    values = c('sienna', 'forestgreen', 'dodgerblue', 'orange
  ) +
  theme_half_open(font_size = 18) +
  theme(legend.position = 'none') +
  labs(
    x = 'Year',
    y = 'Milk produced (billion lbs)',
    title = 'Milk production in four US regions'
  )
```

```
milk_region_anim_plot
```



How to: **Animate a line plot**

Now animate it

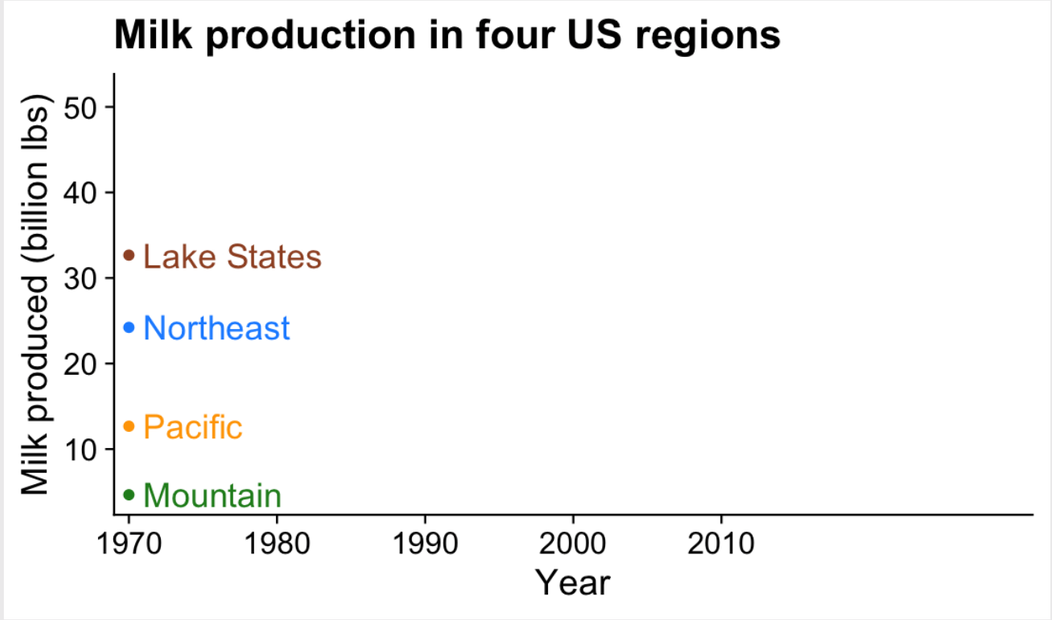
Note the pause at the end!

```
library(gganimate)

milk_region_anim <- milk_region_anim_plot +
    transition_reveal(year)

# Render the animation
animate(
    milk_region_anim,
    end_pause = 15,
    duration = 10,
    width = 1100,
    height = 650,
    res = 150,
    renderer = magick_renderer()
)

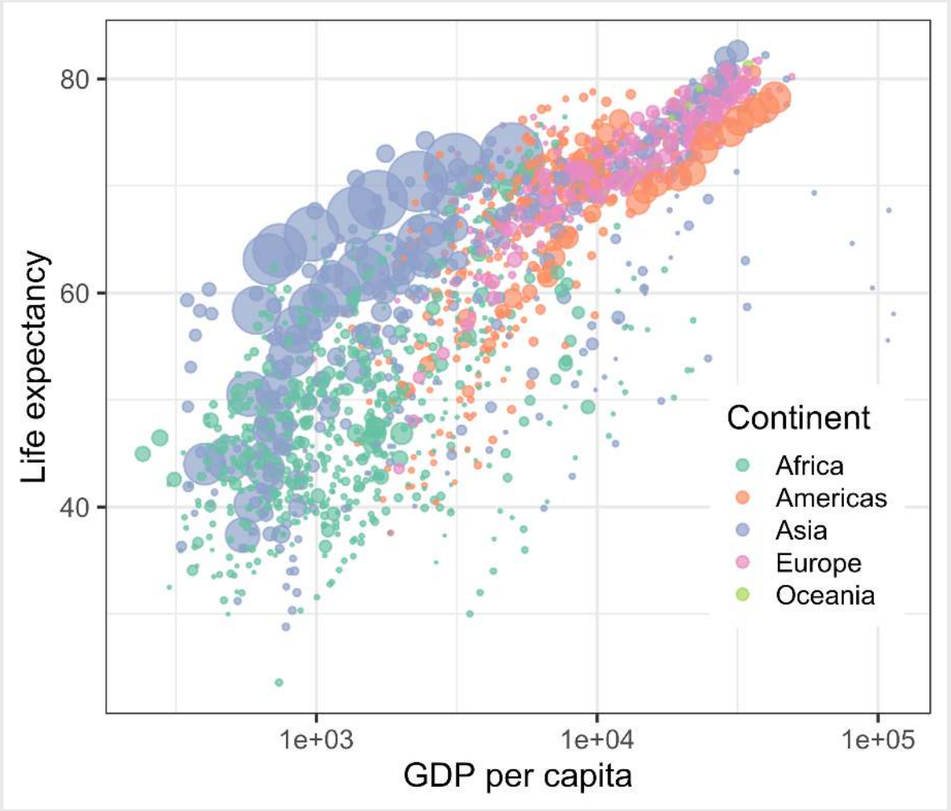
# Save last animation
anim_save(file.path(
    'figs',
    'milk_region_animation.gif'
))
```



How to: **Change label based on year**

First make a static plot

```
 1  gapminder_anim_plot <- ggplot(
 2    gapminder,
 3    aes(x = gdpPercap, y = lifeExp, size = pop, colo
 4  ) +
 5    geom_point(alpha = 0.7) +
 6    scale_size_area(guide = FALSE, max_size = 15) +
 7    scale_color_brewer(palette = 'Set2') +
 8    scale_x_log10() +
 9    theme_bw(base_size = 18) +
10    theme(legend.position = c(0.85, 0.3)) +
11    labs(x = 'GDP per capita', y = 'Life expectancy'
12
13  gapminder_anim_plot
```



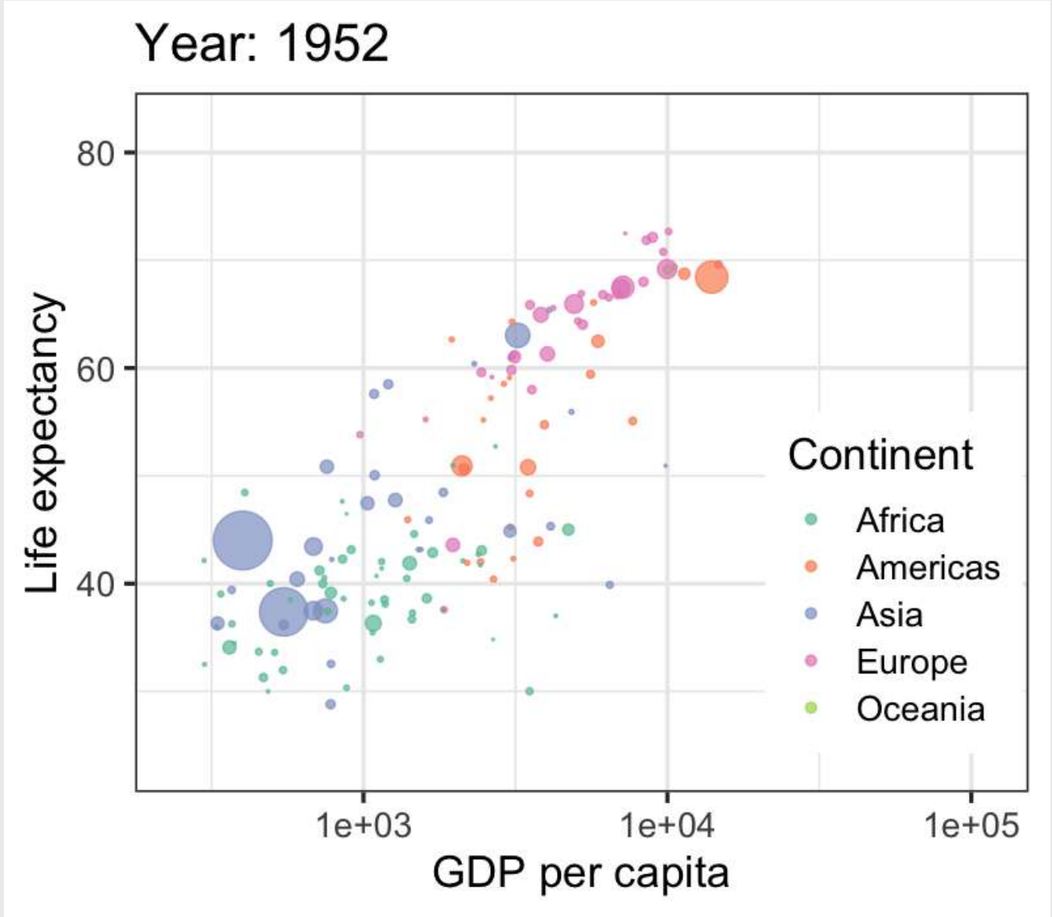
How to: **Change label based on year**

Now animate it

Note: Year must be an integer!

```
gapminder_anim <- gapminder_anim_plot +
  transition_time(year) +
  labs(title = "Year: {frame_time}")

# Render the animation
animate(
  gapminder_anim,
  end_pause = 10,
  width = 800,
  height = 600,
  res = 150,
  renderer = magick_renderer()
)
```



```r
milk_race_anim <- milk_production %>%
  group_by(year) %>%
  mutate(
    rank = rank(-milk_produced),
    Value_rel = milk_produced / milk_produced[rank == 1],
    Value_lbl = paste0(' ', round(milk_produced))
  ) %>%
  group_by(state) %>%
  filter(rank <= 10) %>%
  ungroup() %>%
  mutate(year = as.integer(year)) %>%
  ggplot(aes(x = rank, group = state, fill = region, color = region)) +
  geom_tile(
    aes(y = milk_produced / 2, height = milk_produced),
    width = 0.9,
    alpha = 0.8,
    color = NA
  ) +
  geom_text(aes(y = 0, label = paste(state, " ")), vjust = 0.2, hjust = 1) +
  geom_text(aes(y = milk_produced, label = Value_lbl), hjust = 0) +
  coord_flip(clip = 'off', expand = FALSE) +
  scale_y_continuous(labels = scales::comma) +
  scale_fill_viridis(discrete = TRUE) +
  scale_color_viridis(discrete = TRUE) +
  scale_x_reverse() +
  guides(color = FALSE) +
  theme_minimal_vgrid() +
  transition_time(year) +
  view_follow(fixed_x = TRUE) +
  theme(
    axis.line = element_blank(),
    axis.text = element_blank(),
    axis.ticks = element_blank(),
    axis.title = element_blank(),
    legend.position = c(0.7, 0.3),
    legend.background = element_rect(fill = 'white'),
    plot.title = element_text(
      size = 22,
      hjust = 0.5,
      face = 'bold',
      colour = 'grey',
      vjust = -1
    ),
    plot.subtitle = element_text(
      size = 18,
      hjust = 0.5,
      face = 'italic',
      color = 'grey'
    ),
    plot.caption = element_text(
      size = 8,
      hjust = 0.5,
```

Making a bar chart race (tutorial here)

```r
animate(
        milk_race_anim,
        duration = 17,
        end_pause = 15,
        width = 800,
        height = 700,
        res = 150,
        renderer = magick_renderer()
)
```

Resources

More animation options:

- More on gapminder + line charts
- Customizing the animation

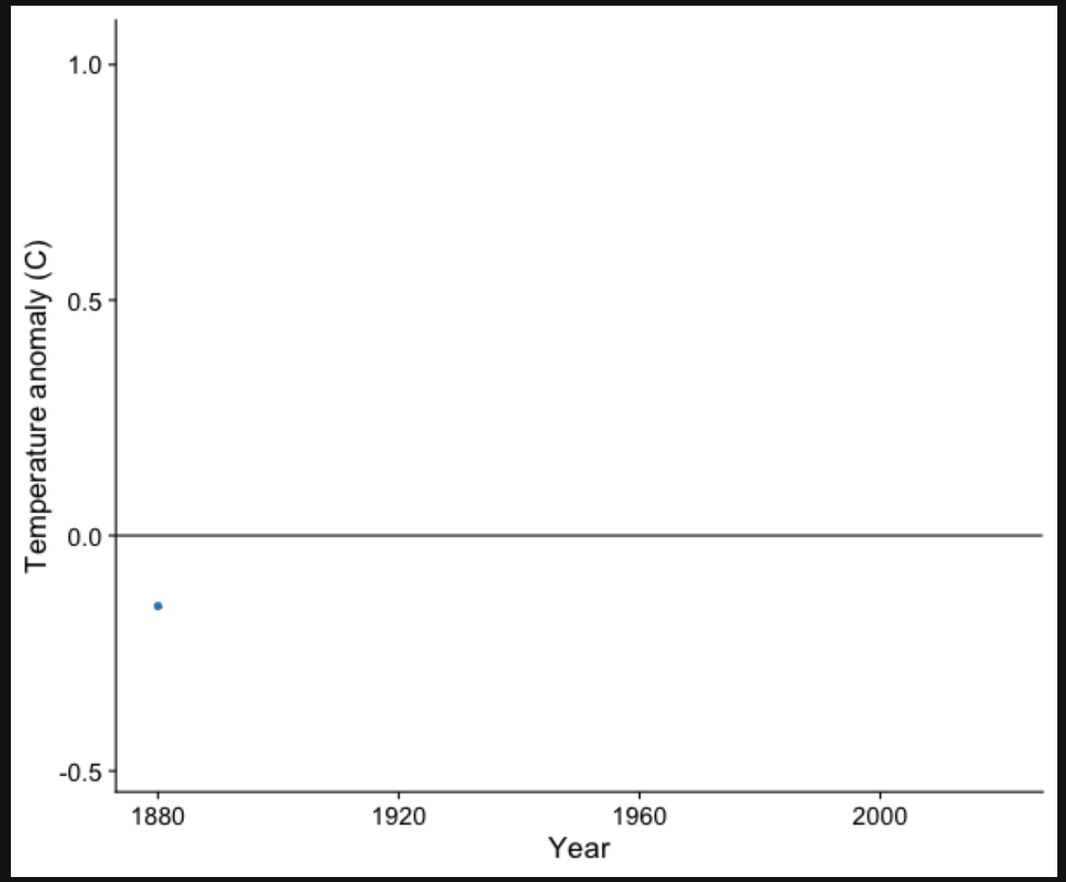
Your turn

15:00

Use the `global_temps` data frame to explore ways to *animate* the change in average global temperatures.

Consider using:

- points
- lines
- areas



Break!

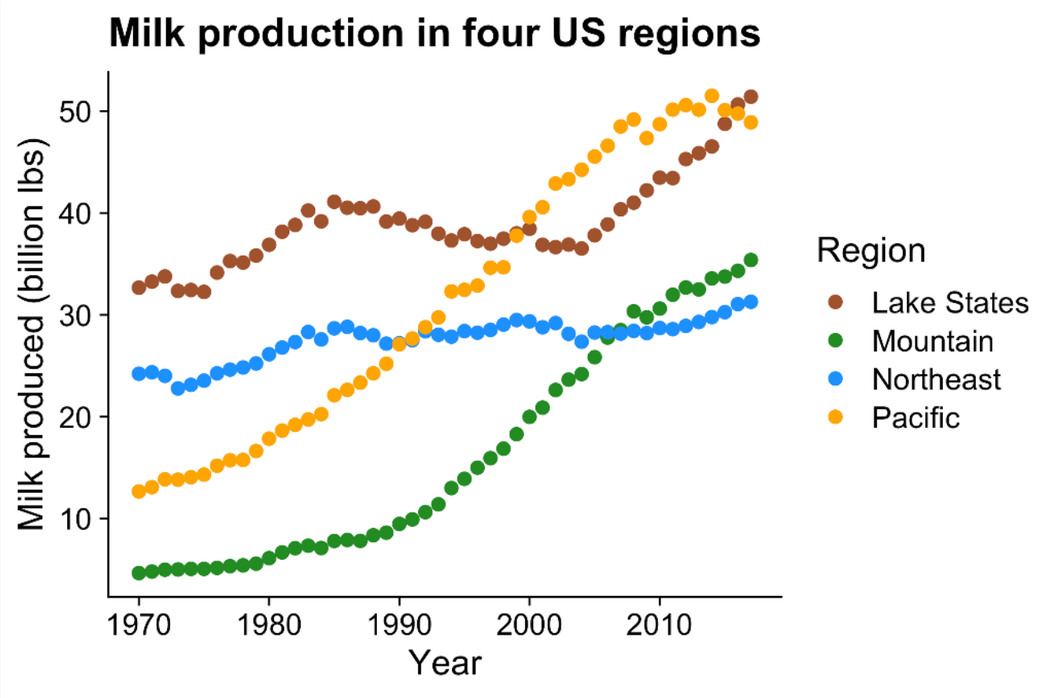
Stand up, Move around, Stretch!

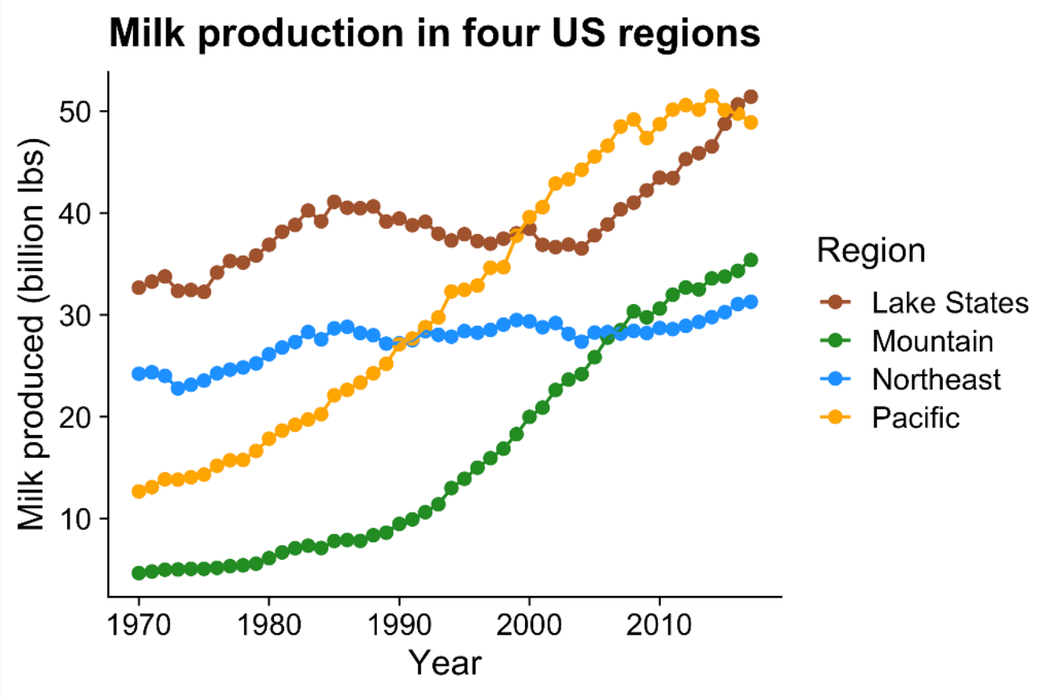
05:00

Week 10: *Trends*

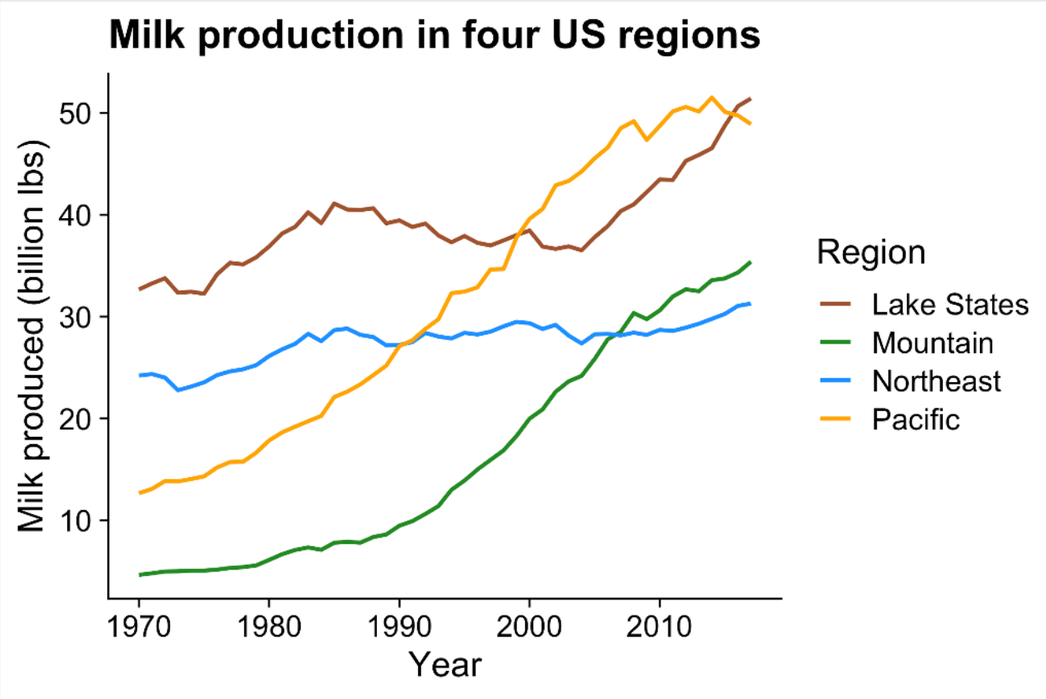
1. Single Variables
2. Animations
3. Multiple Variables

With multiple categories, points & lines can get messy

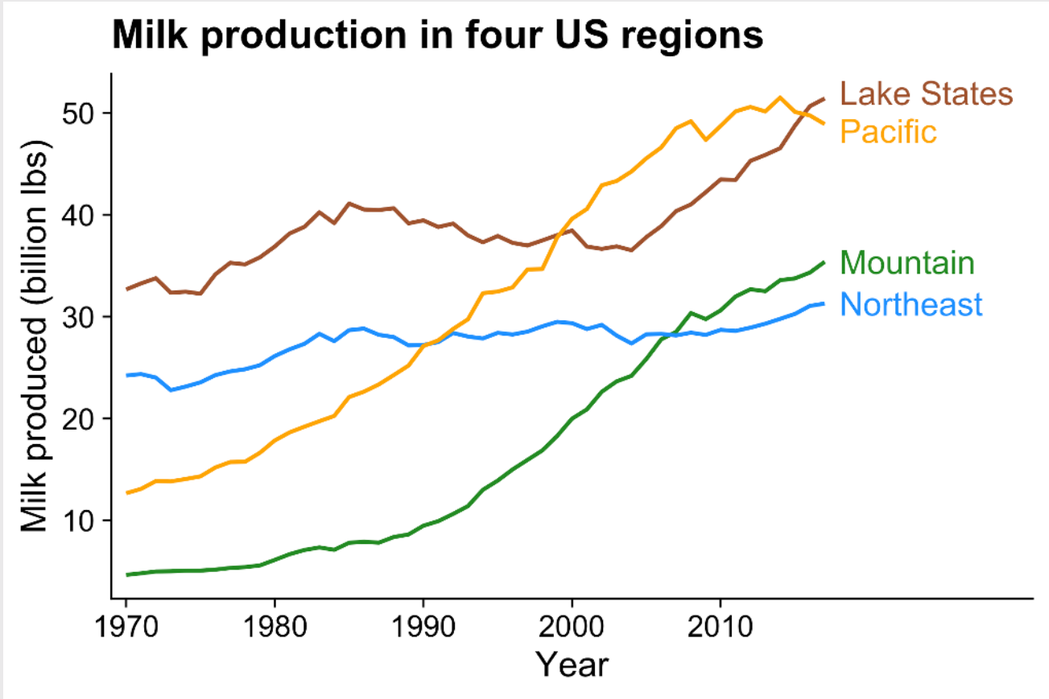




Better: Lines alone makes distinguishing trends easier

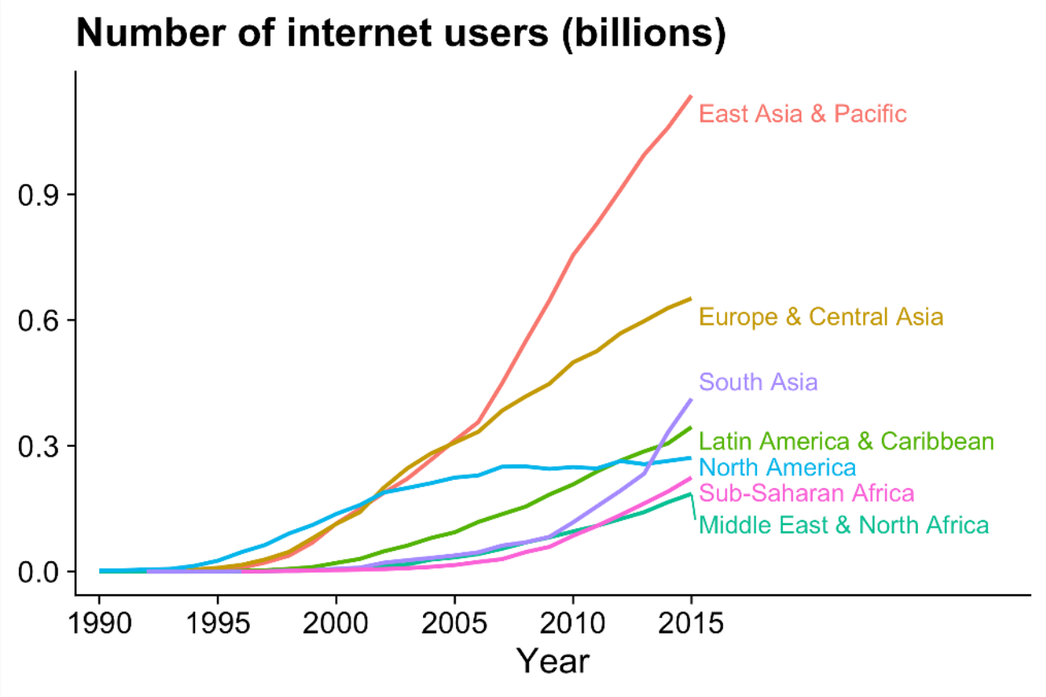


Even better: Directly label lines to remove legend

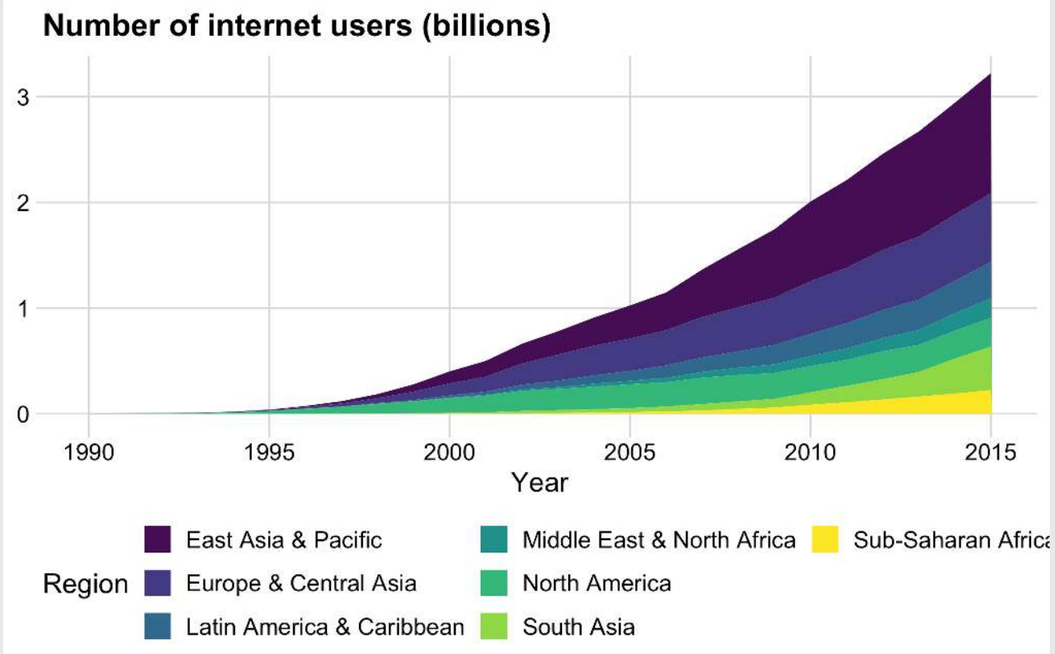


If goal is to communicate the **overall / total** trend, consider a stacked area chart

Highlights **regional** trends



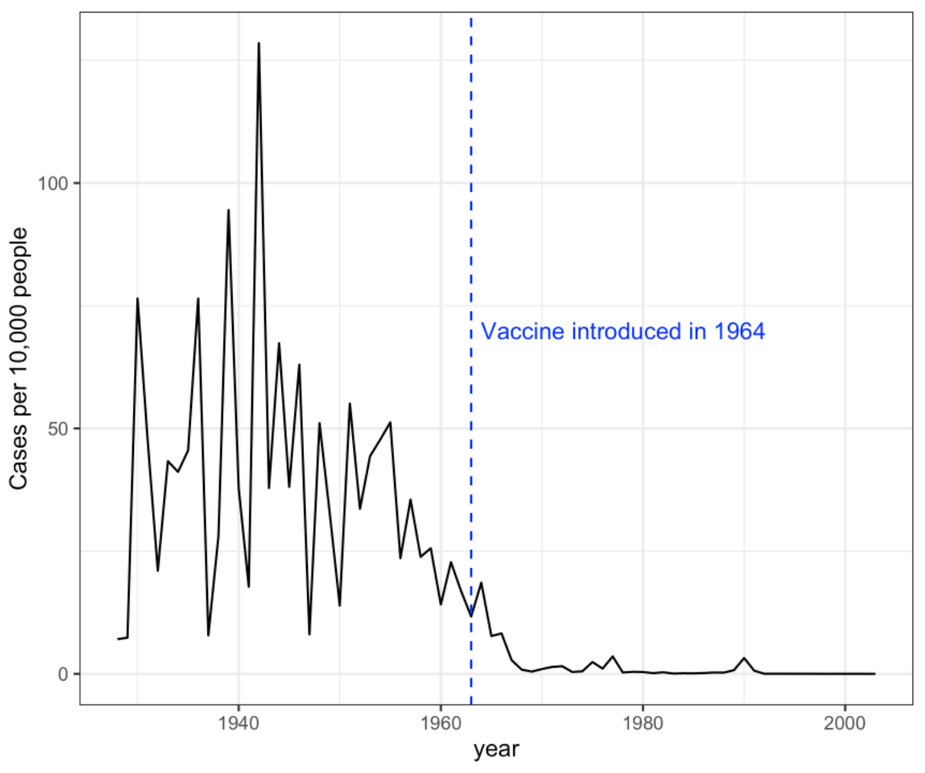
Highlights **overall / total** trend



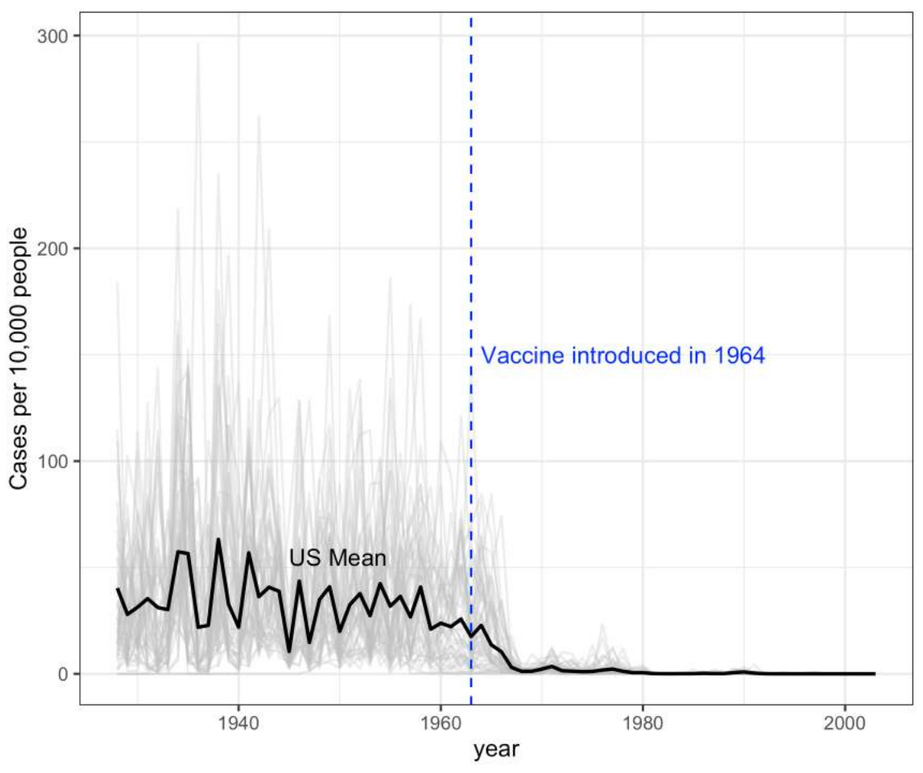
If you have **lots** of categories:

1) Plot all the data with the average highlighted

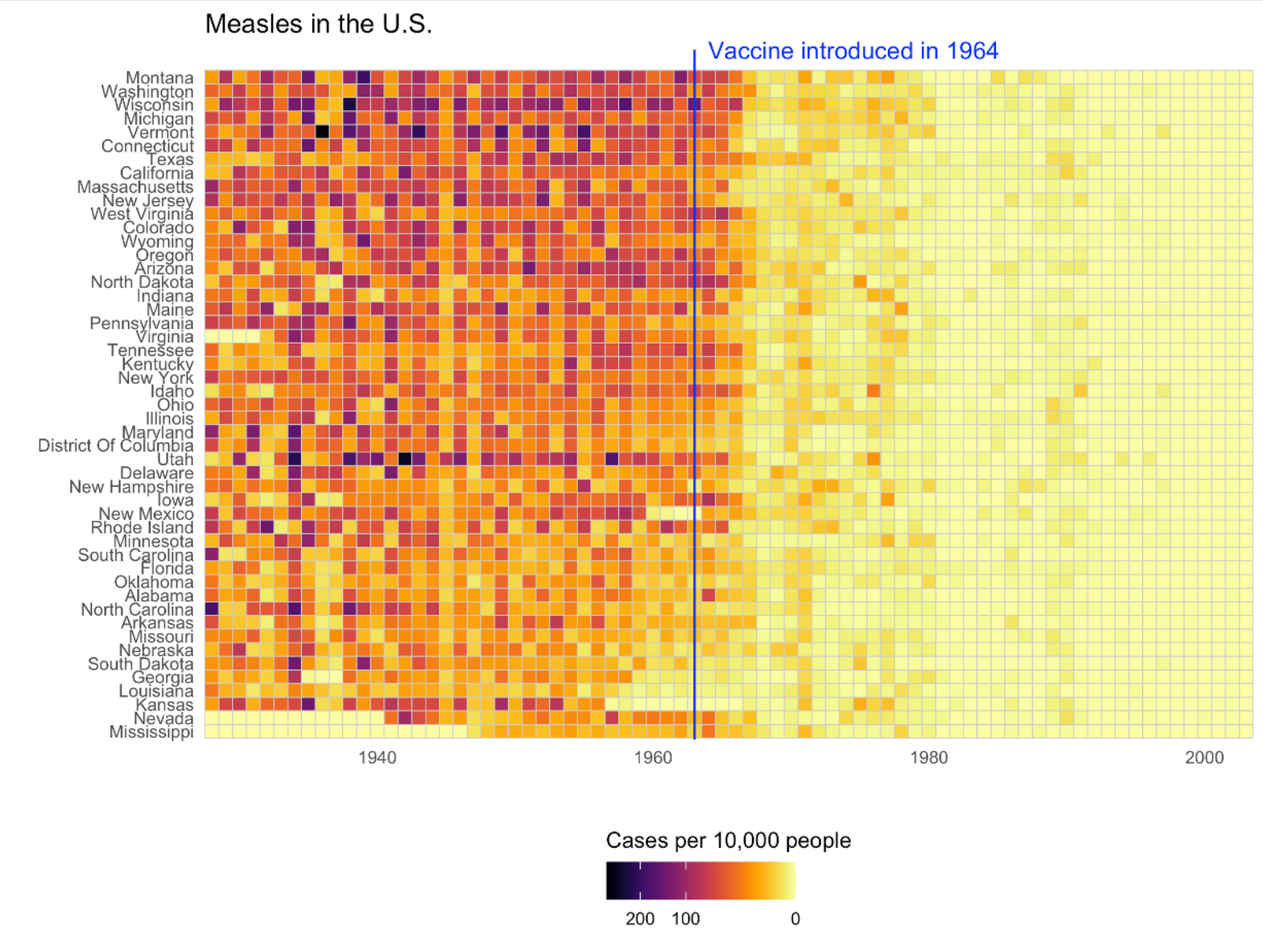
Measles in **California**



Measles in **all 50 states**

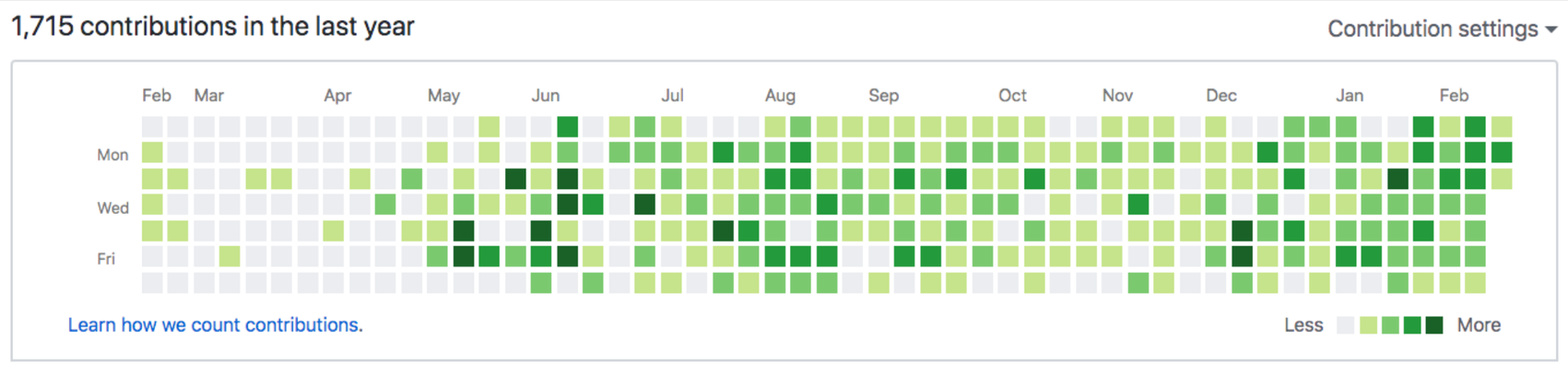


If you have **lots** of categories:
1) Plot all the data with the average highlighted
2) Plot all the data with a heat map



Heatmaps are great for multiple divisions of time

My activity on Github:



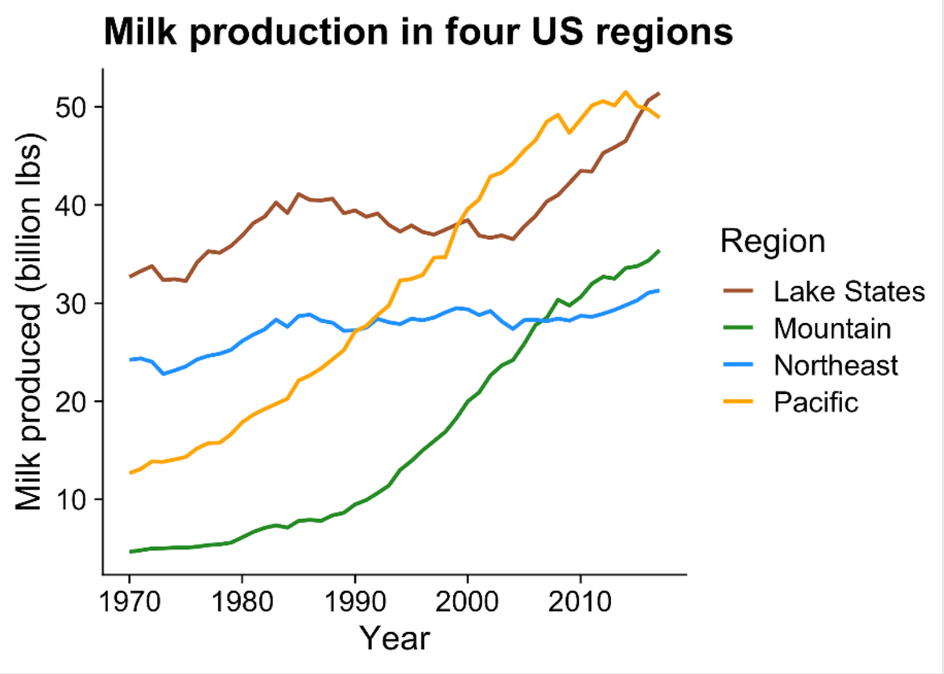
Check out this heat map on Traffic fatalities

Make the basic line chart first

```r
# Format the data
milk_region <- milk_production %>%
  filter(
    region %in%
      c('Pacific', 'Northeast', 'Lake States', 'Mountain')
  ) %>%
  group_by(year, region) %>%
  summarise(milk_produced = sum(milk_produced)) %>%
  ungroup()

# Make the line chart
ggplot(
  milk_region,
  aes(x = year, y = milk_produced, color = region)
) +
  geom_line(size = 1) +
  scale_color_manual(
    values = c('sienna', 'forestgreen', 'dodgerblue', 'orange')
  ) +
  theme_half_open(font_size = 18) +
  labs(
    x = 'Year',
    y = 'Milk produced (billion lbs)',
    color = 'Region',
    title = 'Milk production in four US regions'
  )
```

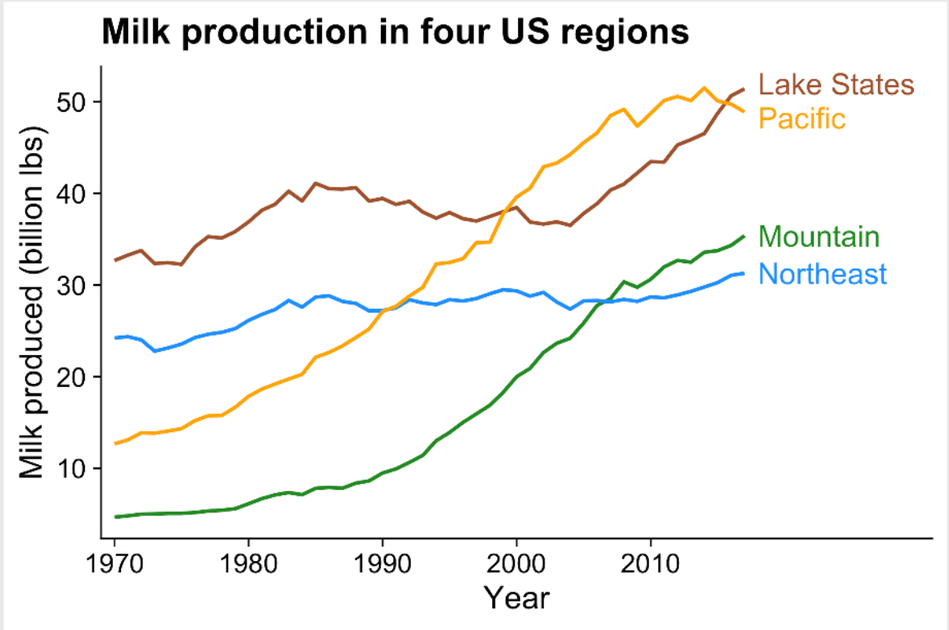
How to:
Directly label lines



How to:
Directly label lines

```r
# Format the data
milk_region <- milk_production %>%
  filter(
    region %in%
      c('Pacific', 'Northeast', 'Lake States', 'Mountain')
  ) %>%
  group_by(year, region) %>%
  summarise(milk_produced = sum(milk_produced)) %>%
  ungroup()

# Make the line plot
ggplot(milk_region, aes(x = year, y = milk_produced, color = region
  geom_line(size = 1) +
  # Add labels
  geom_text_repel(
    data = milk_region %>% filter(year == max(year)),
    aes(label = region),
    hjust = 0,
    nudge_x = 1,
    direction = "y",
    size = 6,
    segment.color = NA
  ) +
  # Create space for labels on right side
  scale_x_continuous(
    breaks = seq(1970, 2010, 10),
    expand = expansion(add = c(1, 13))
  ) +
  scale_color_manual(
    values = c('sienna', 'forestgreen', 'dodgerblue', 'orange')
  ) +
  theme_half_open(font_size = 18) +
  # Remove legend
  theme(legend.position = 'none') +
  labs(
```



```r
library(geomtextpath)

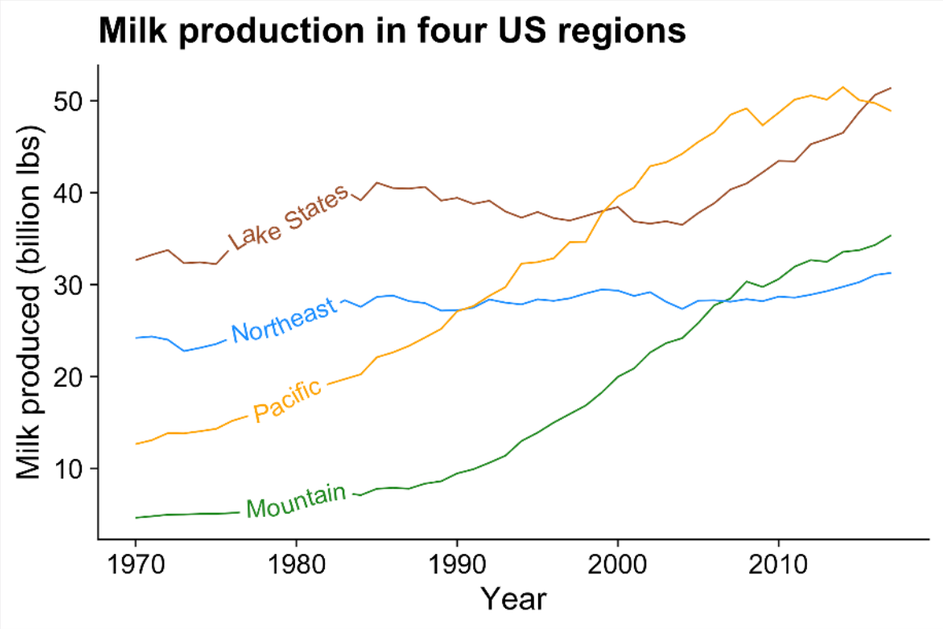
# Format the data
milk_production %>%
  filter(
    region %in%
      c('Pacific', 'Northeast', 'Lake States', 'Mountain')
  ) %>%
  group_by(year, region) %>%
  summarise(milk_produced = sum(milk_produced)) %>%
  ungroup() %>%

  # Make the line plot
  ggplot() +
  geom_textline(
    aes(
      x = year,
      y = milk_produced,
      color = region,
      label = region,
      group = region
    ),
    size = 5,
    hjust = 0.15
  ) +
  scale_color_manual(
    values = c('sienna', 'forestgreen', 'dodgerblue', 'orange')
  ) +
  theme_half_open(font_size = 18) +
  # Remove legend
  theme(legend.position = 'none') +
  labs(
    x = 'Year',
    y = 'Milk produced (billion lbs)',
    title = 'Milk production in four US regions'
```

Alternative: **Embed the labels!**

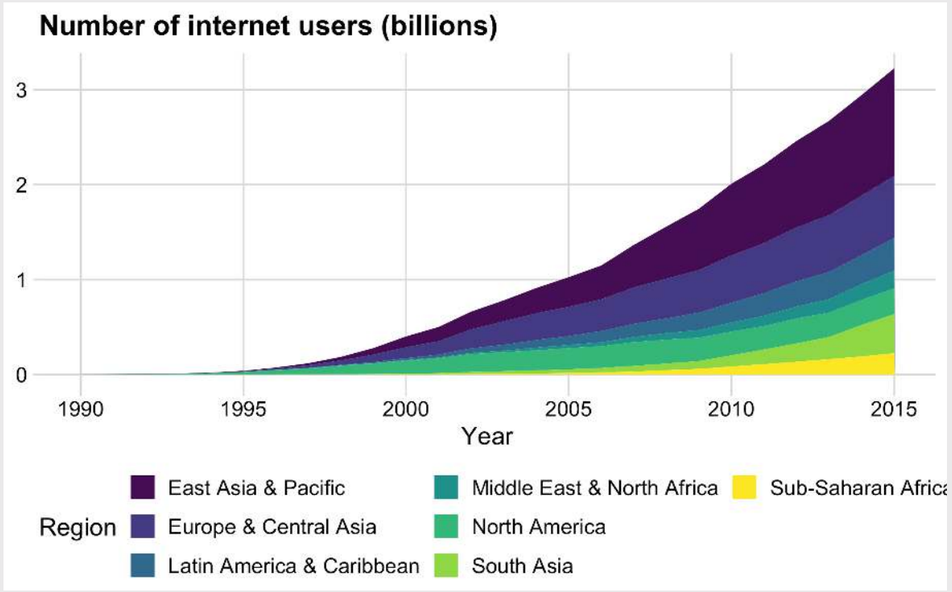
Use `{geomtextpath}` package

(see this SO issue for other strategies)



How to: **Stacked area**

```r
internet_region %>%
  mutate(numUsers = numUsers / 10^9) %>%
  ggplot() +
  geom_area(aes(x = year, y = numUsers, fill = region)) +
  # Nice colors from "viridis" library:
  scale_fill_viridis(discrete = TRUE) +
  # Sort the legend into 3 rows
  guides(
    fill = guide_legend(
      nrow = 3,
      byrow = FALSE
    )
  ) +
  theme_minimal_grid(font_size = 15) +
  theme(legend.position = 'bottom') +
  labs(
    x = 'Year',
    y = NULL,
    fill = 'Region',
    title = 'Number of internet users (billions)'
  )
```



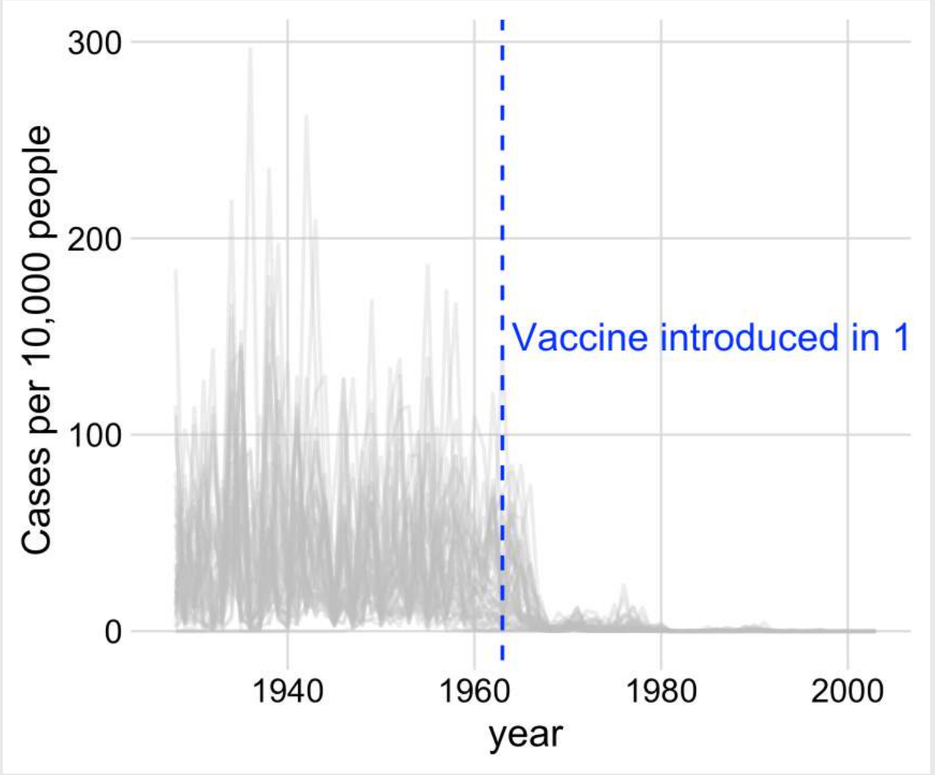
Format the data

```r
# Format the data
measles <- us_diseases %>%
  filter(
    disease == 'Measles',
    !state %in% c("Hawaii", "Alaska")
  ) %>%
  mutate(
    rate = (count / population) * 10000,
    state = fct_reorder(state, rate)
  ) %>%
  # Compute annual mean rate across all states
  group_by(year) %>%
  mutate(
    mean_rate = sum(count) / sum(population) * 10000
  )
```

Make all the state lines in light grey color

```r
ggplot(measles) +
    geom_line(
        aes(x = year, y = rate, group = state),
        color = 'grey',
        alpha = 0.3
    ) +
    # Add reference line & label:
    geom_vline(xintercept = 1963, col = 'blue', linetype = 'd
    annotate(
        'text',
        x = 1964,
        y = 150,
        hjust = 0,
        label = 'Vaccine introduced in 1964',
```

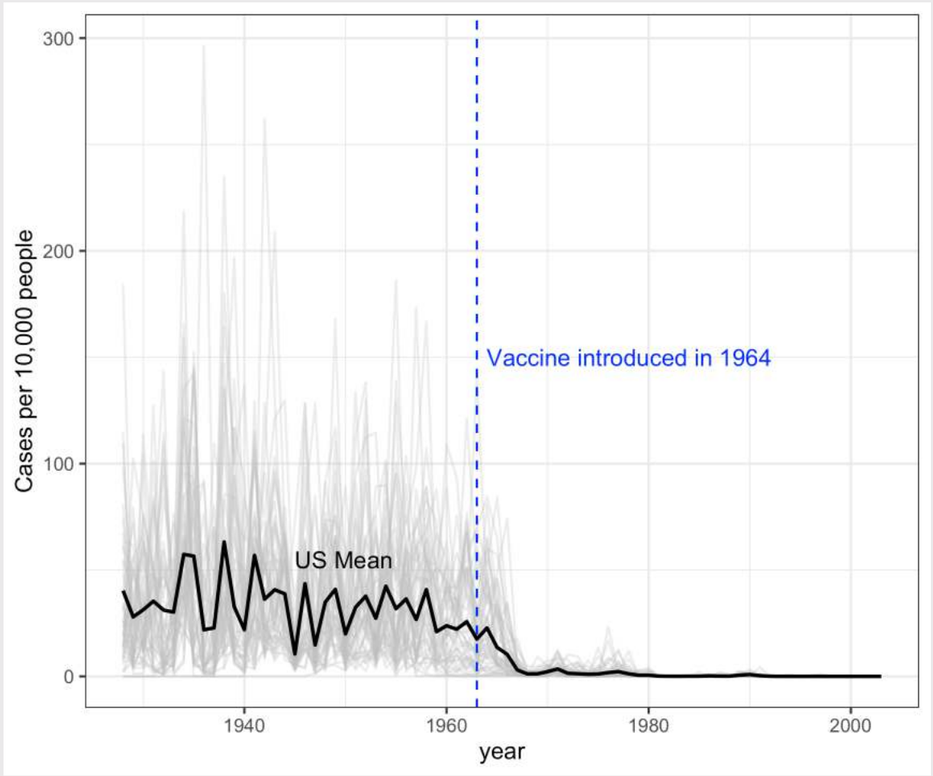
How to:
Average line overlay



Now overlay the annual mean line

```
ggplot(measles) +
  geom_line(
    aes(x = year, y = rate, group = state),
    color = 'grey',
    alpha = 0.3
  ) +
  geom_line(
    aes(x = year, y = mean_rate),
    size = 0.8
  ) +
  # Add US mean label
  annotate(
    'text',
    x = 1945,
    y = 55,
    hjust = 0,
    label = 'US Mean'
  ) +
  # Add reference line & label
  geom_vline(xintercept = 1963, col = 'blue', linetype =
  annotate(
    'text',
    x = 1964,
    y = 150,
    hjust = 0,
    label = 'Vaccine introduced in 1964',
    color = 'blue'
  ) +
```

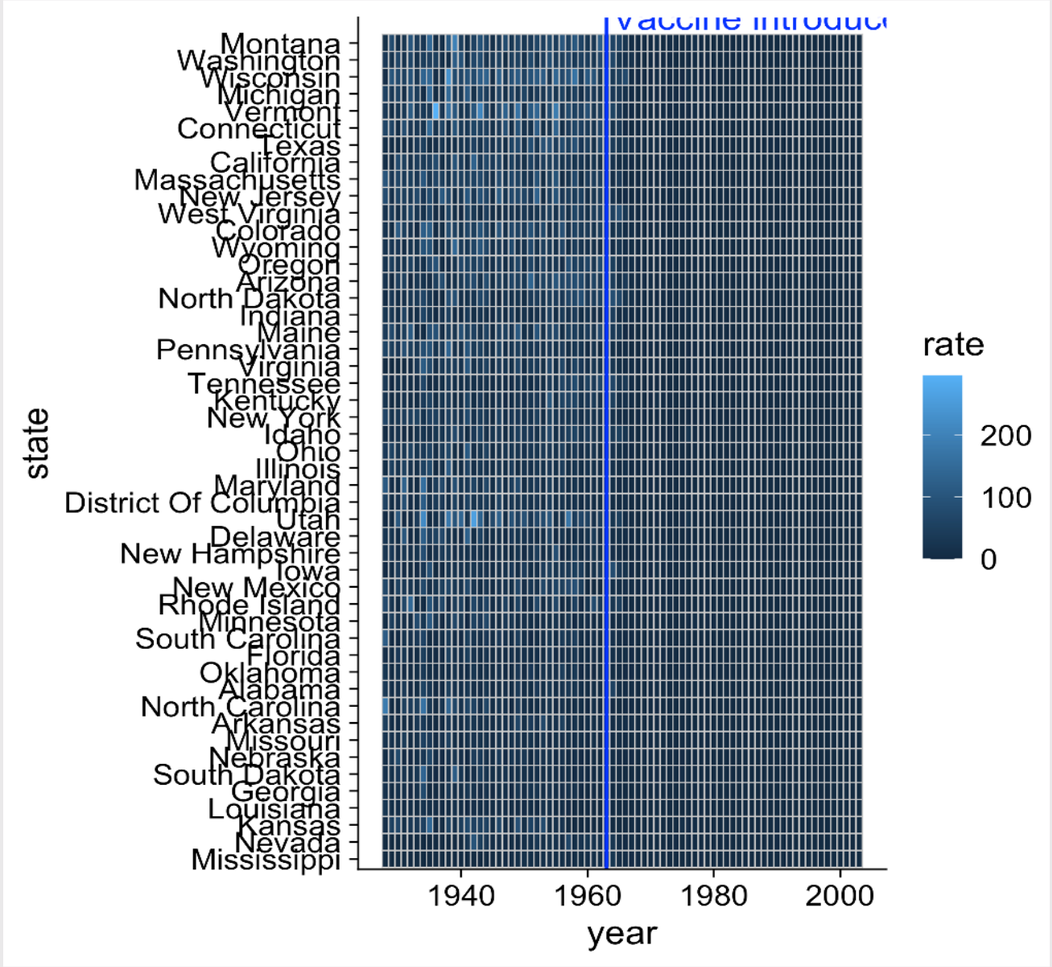
How to:
Average line overlay



How to: **Heat map**

Create main grid with `geom_tile()`

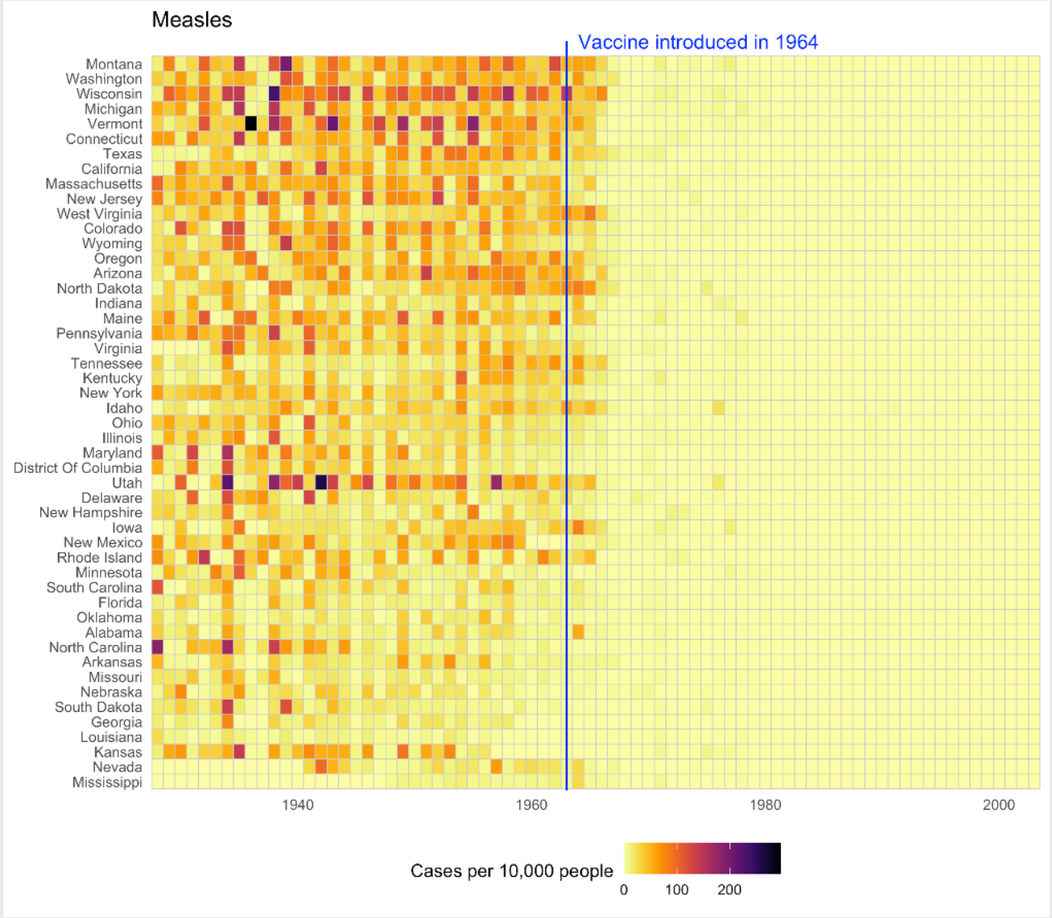
```
ggplot(measles) +
  geom_tile(
    aes(x = year, y = state, fill = rate),
    color = 'grey80'
  ) +
  # Add reference line & label
  geom_vline(
    xintercept = 1963,
    col = 'blue'
  ) +
  annotate(
    'text',
    x = 1964,
    y = 50.5,
    hjust = 0,
    label = 'Vaccine introduced in 1964',
    color = 'blue'
  )
```



Adjust scales and adjust theme

```r
ggplot(measles) +
  geom_tile(aes(x = year, y = state, fill = rate), color = 'g
  # Add reference line & label
  geom_vline(xintercept = 1963, col = 'blue') +
  annotate(
    'text',
    x = 1964,
    y = 50.5,
    hjust = 0,
    label = 'Vaccine introduced in 1964',
    color = 'blue'
  ) +
  # Adjust scales
  scale_x_continuous(expand = c(0, 0)) +
  scale_fill_viridis(
    option = 'inferno',
    direction = -1
  ) +
  # Adjust theme
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    legend.position = 'bottom',
    text = element_text(size = 10)
  ) +
  coord_cartesian(clip = 'off') +
  labs(
    x = NULL,
    y = NULL,
    fill = 'Cases per 10,000 people',
    title = 'Measles'
  )
```

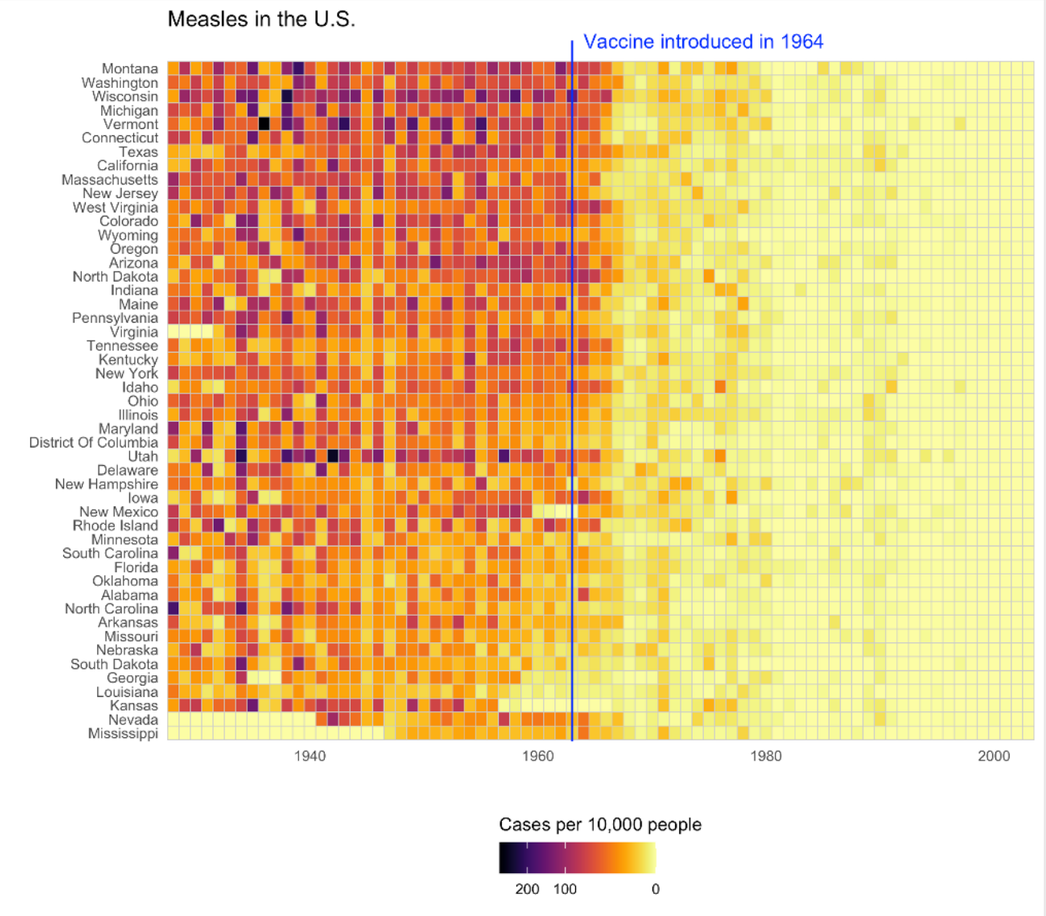
Color scale is linear in this chart



Adjust scales and adjust theme

```
ggplot(measles) +
  geom_tile(aes(x = year, y = state, fill = rate), color = 'g
  # Add reference line & label
  geom_vline(xintercept = 1963, col = 'blue') +
  annotate(
    'text',
    x = 1964,
    y = 50.5,
    hjust = 0,
    label = 'Vaccine introduced in 1964',
    color = 'blue'
  ) +
  # Adjust scales
  scale_x_continuous(expand = c(0, 0)) +
  scale_fill_viridis(
    option = 'inferno',
    direction = -1,
    trans = 'sqrt'
  ) +
  # Modify legend color bar
  guides(
    fill = guide_colorbar(
      title.position = 'top',
      reverse = TRUE
    )
  ) +
  # Adjust theme
  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    legend.position = 'bottom',
    text = element_text(size = 10)
```

Non-linear color scale helps with large variations



Your turn

20:00

Use the `us_covid` data frame to explore ways to visualize the number of daily cases using:

1. A labeled line chart
2. A stacked area chart
3. A heat map

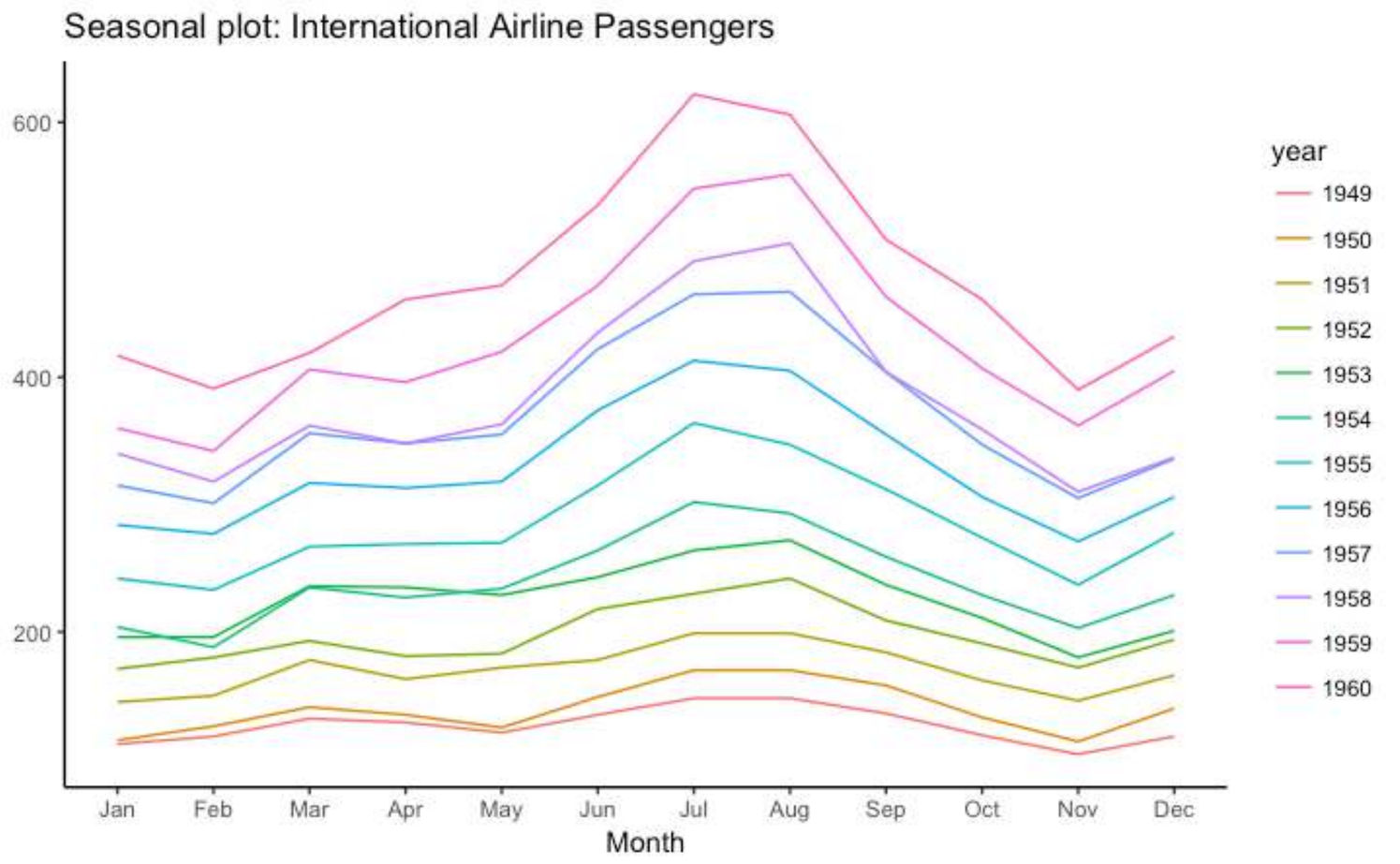
```
us_covid <- read_csv(file.path(
  'data',
  'us_covid.csv'
))

head(us_covid)
```

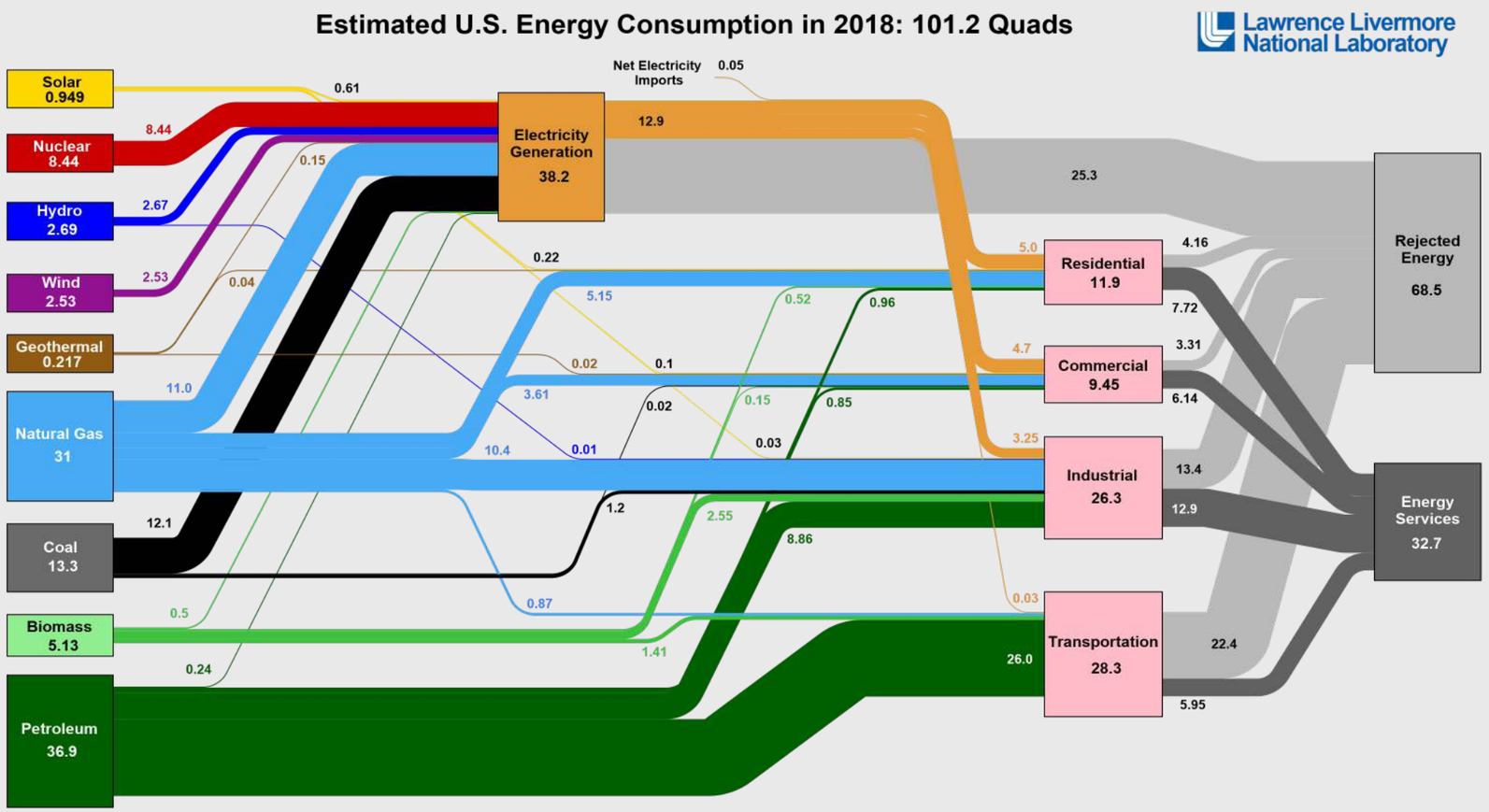
```
#> # A tibble: 6 × 7
#>   date         day state   cases_daily deaths_daily c
#>   <date>     <dbl> <chr>         <dbl>        <dbl>
#> 1 2020-01-23     1 Alabama           0            0
#> 2 2020-01-24     2 Alabama           0            0
#> 3 2020-01-25     3 Alabama           0            0
#> 4 2020-01-26     4 Alabama           0            0
#> 5 2020-01-27     5 Alabama           0            0
#> 6 2020-01-28     6 Alabama           0            0
```

Two other examples for showing change across multiple categories

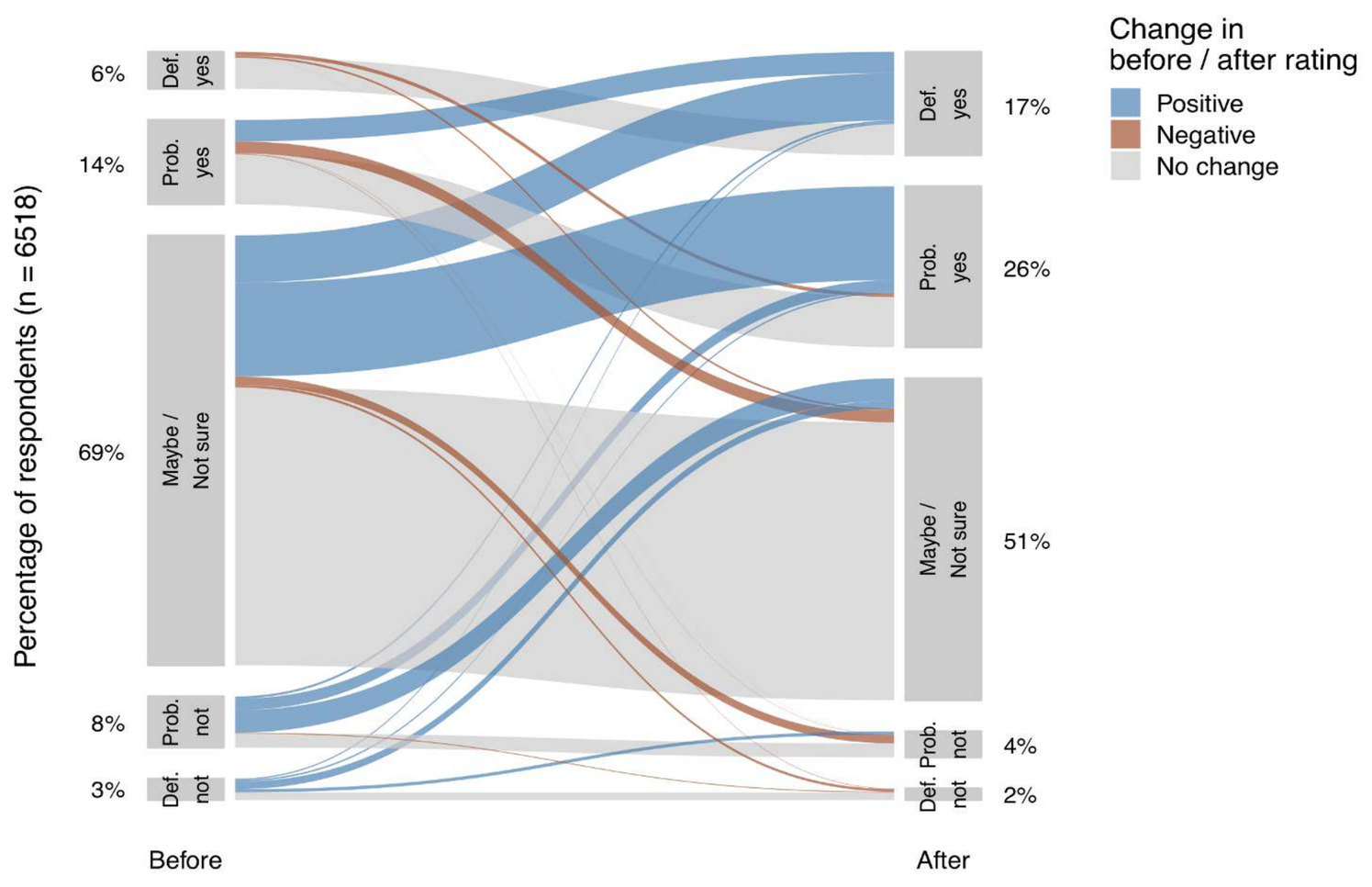
Seasonal chart



Sankey chart



Would you consider purchasing an electric car?



Roberson, Laura A. & Helveston, J.P. (2020) "Electric vehicle adoption: can short experiences lead to big change?," Environmental Research Letters. 15(0940c3).
Made using the ggforce package